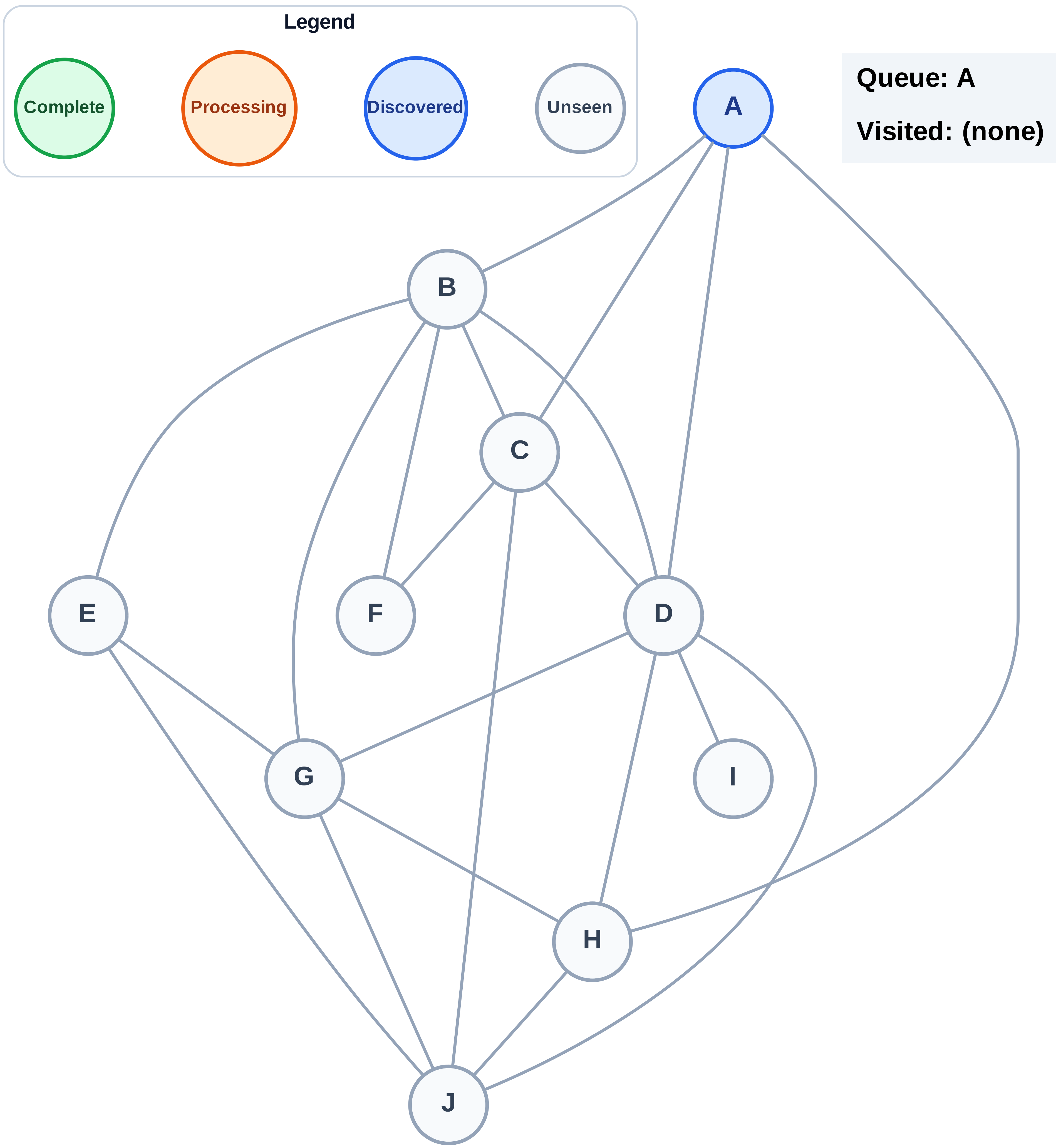
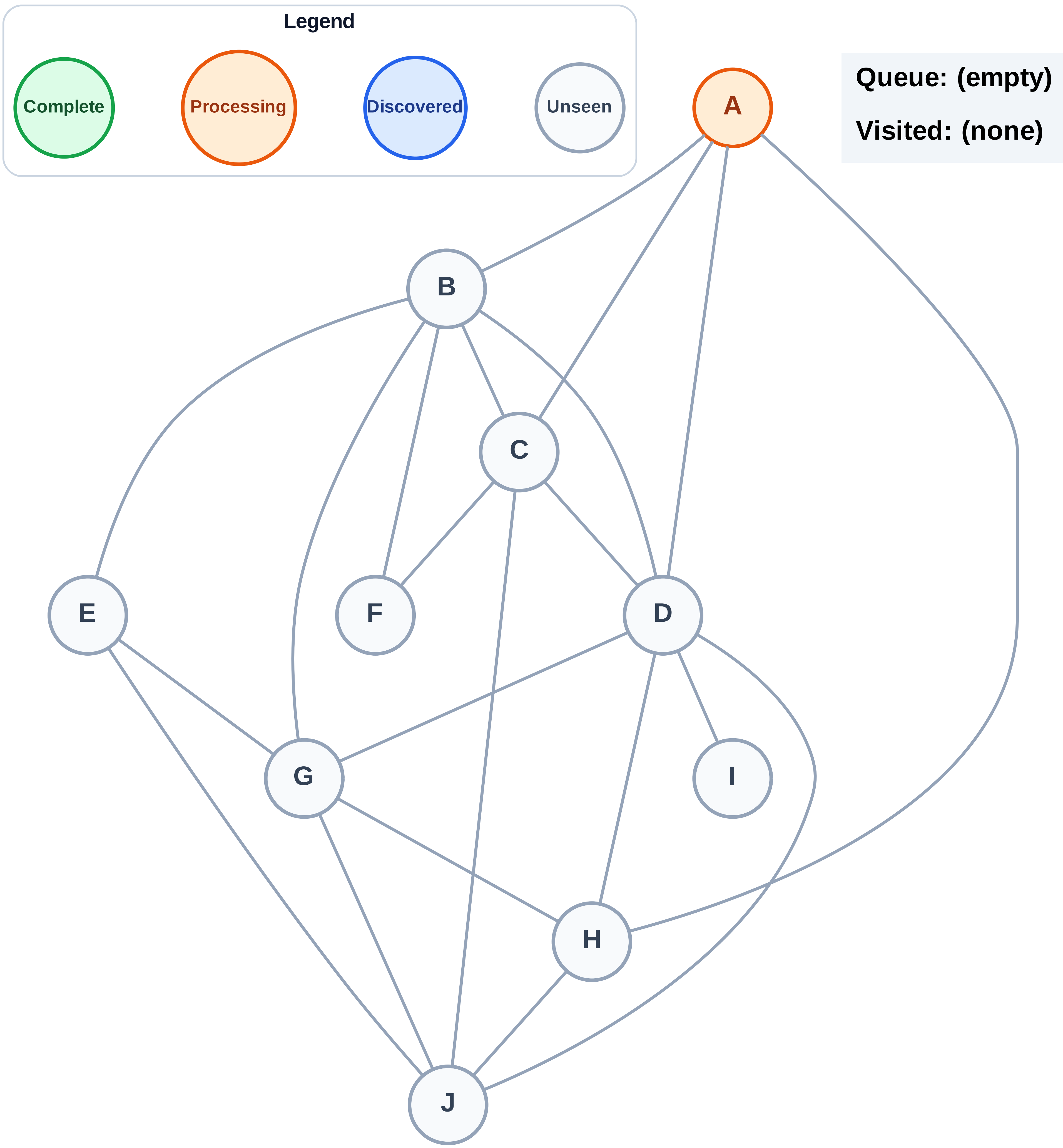
BFS Traversal

Step 1: Start at A



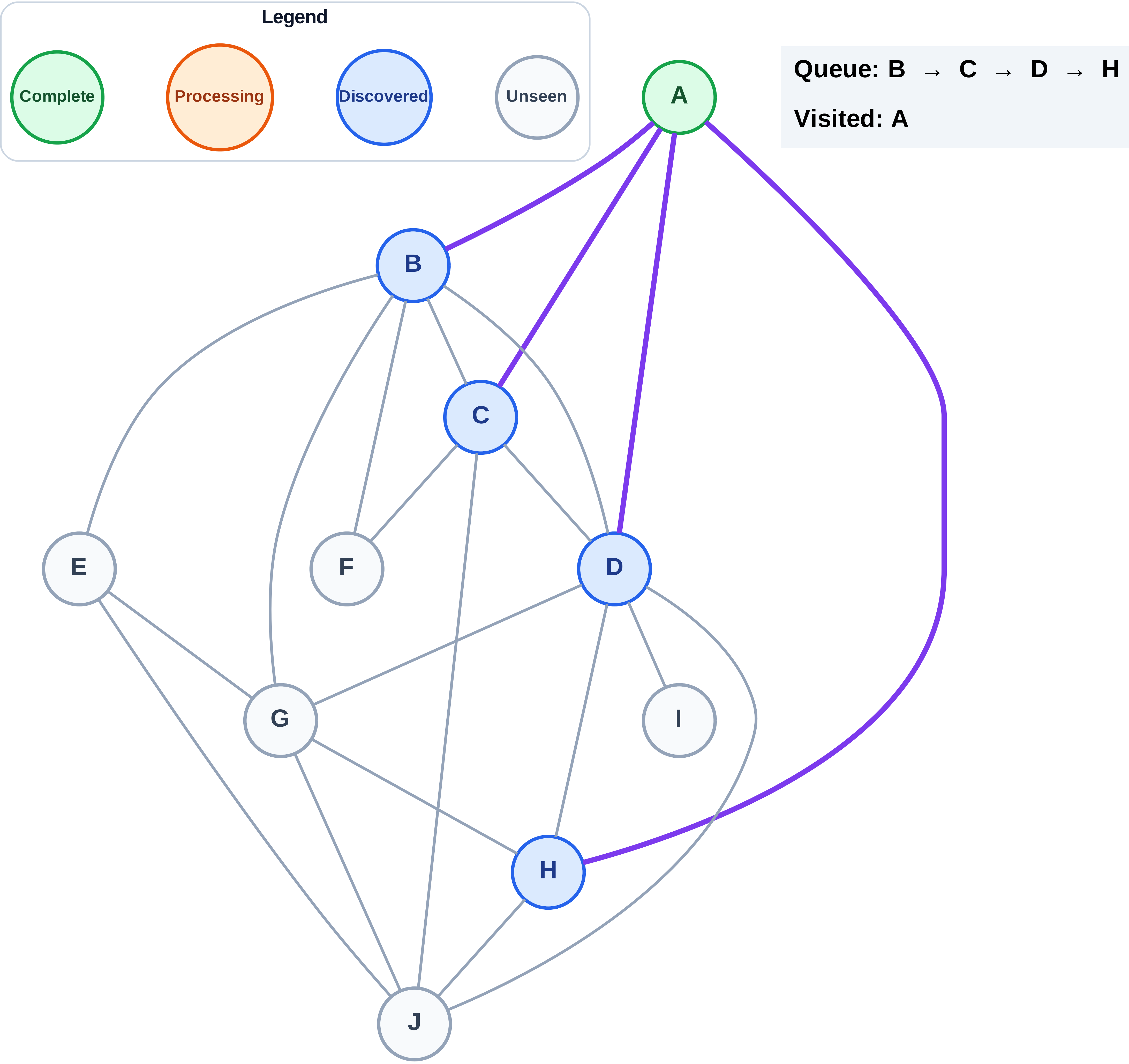
BFS Traversal

Step 2: Process node A



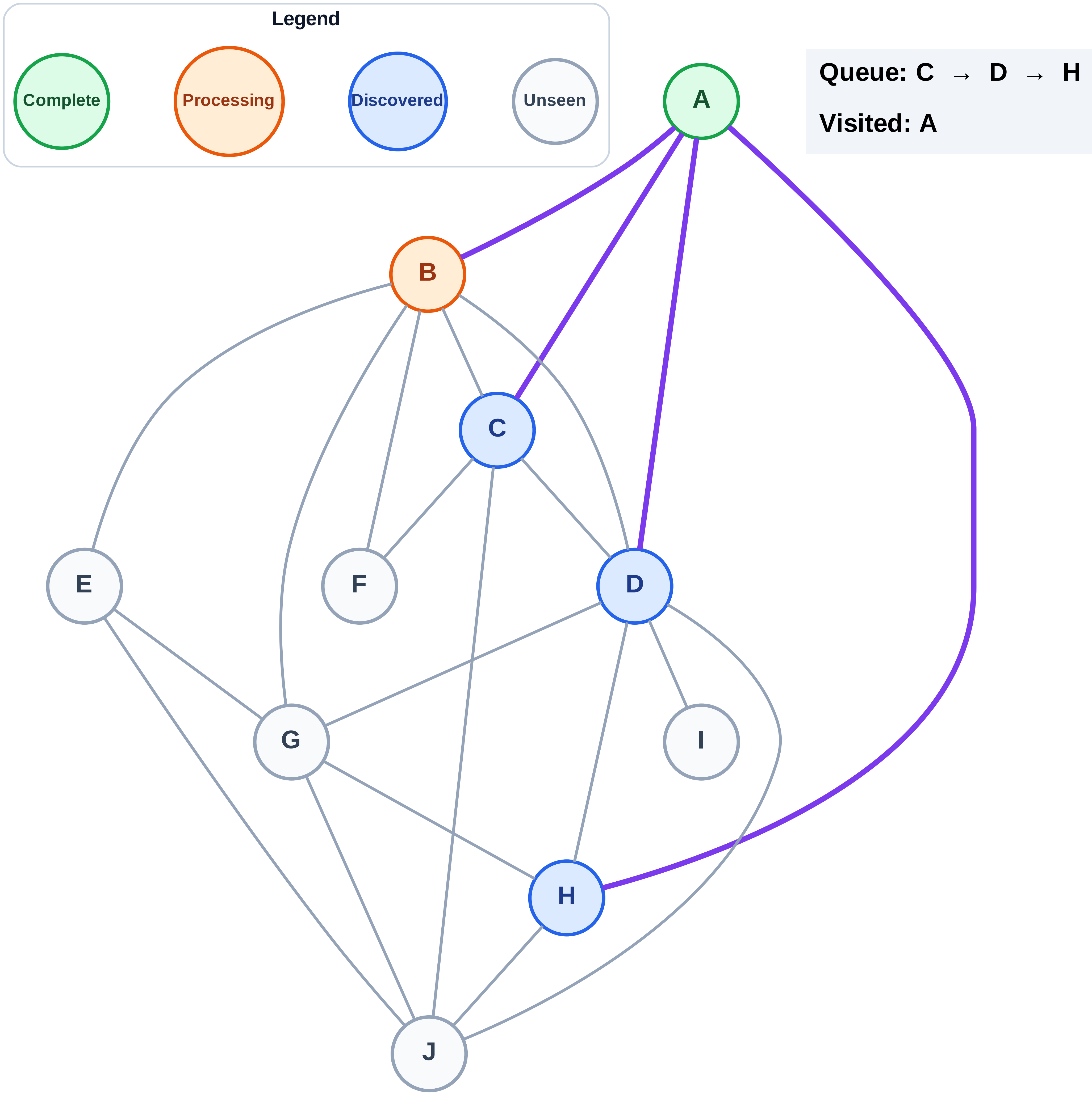
BFS Traversal

Step 3: Discover B, C, D, H from A



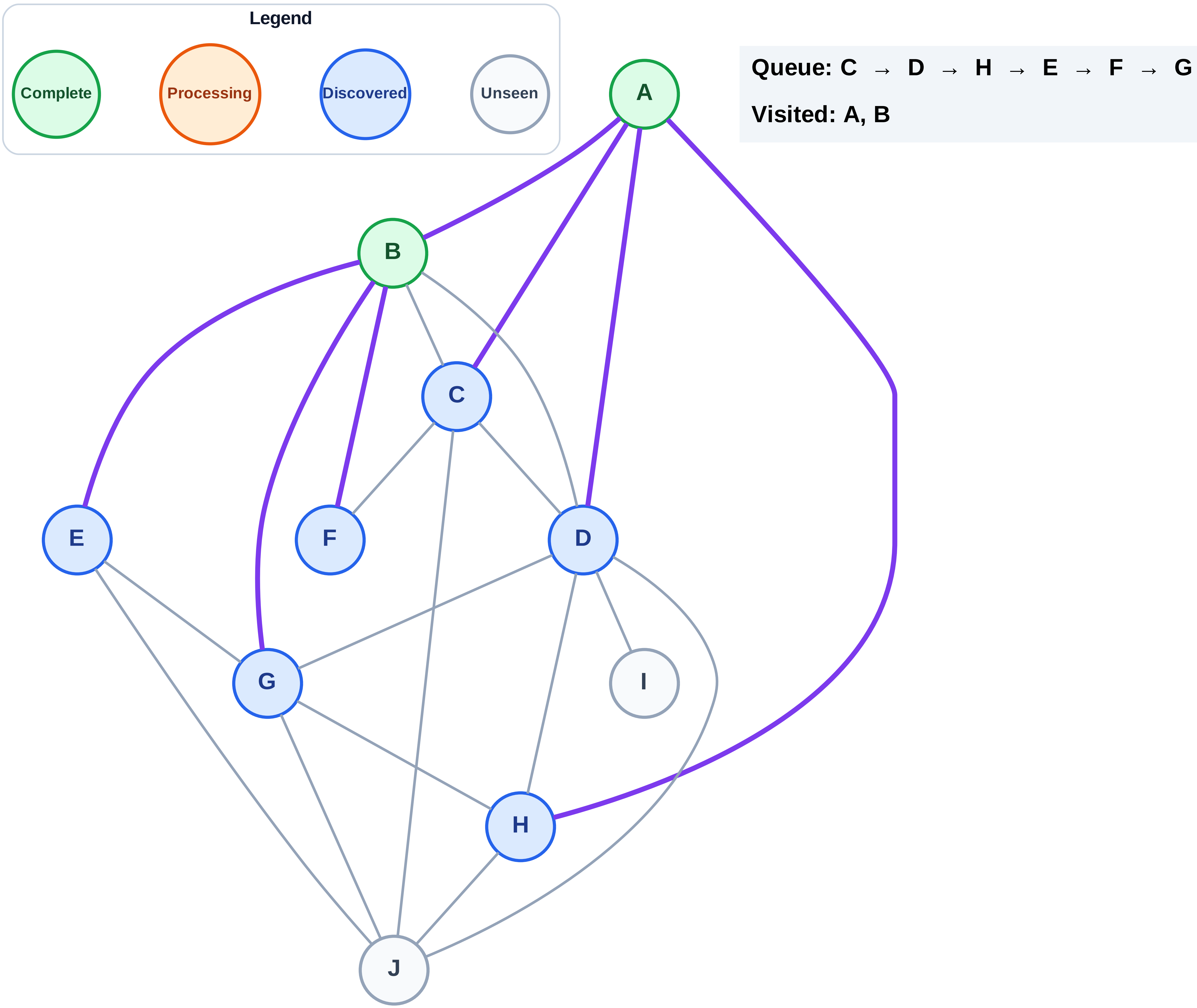
BFS Traversal

Step 4: Process node B



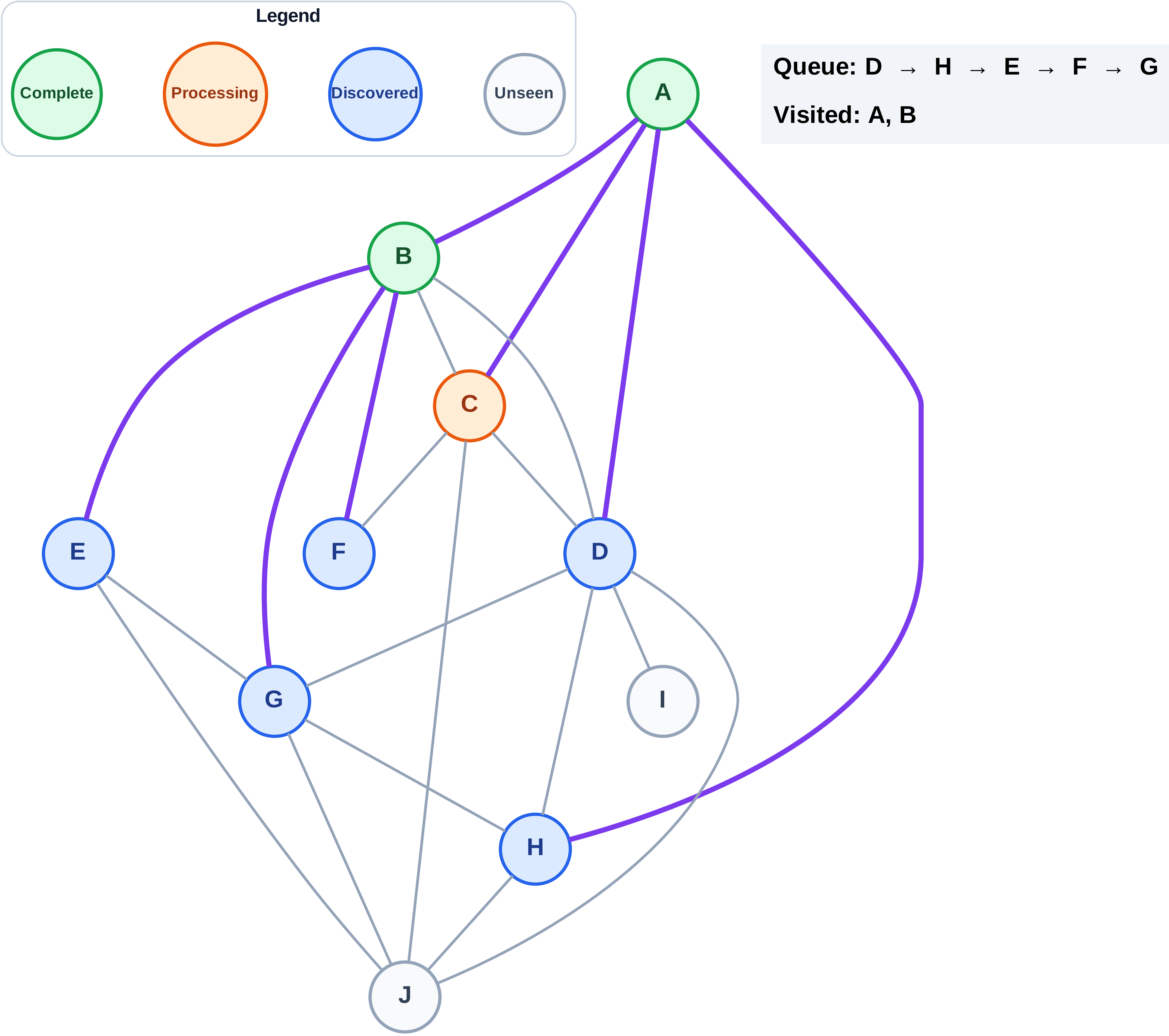
BFS Traversal

Step 5: Discover E, F, G from B



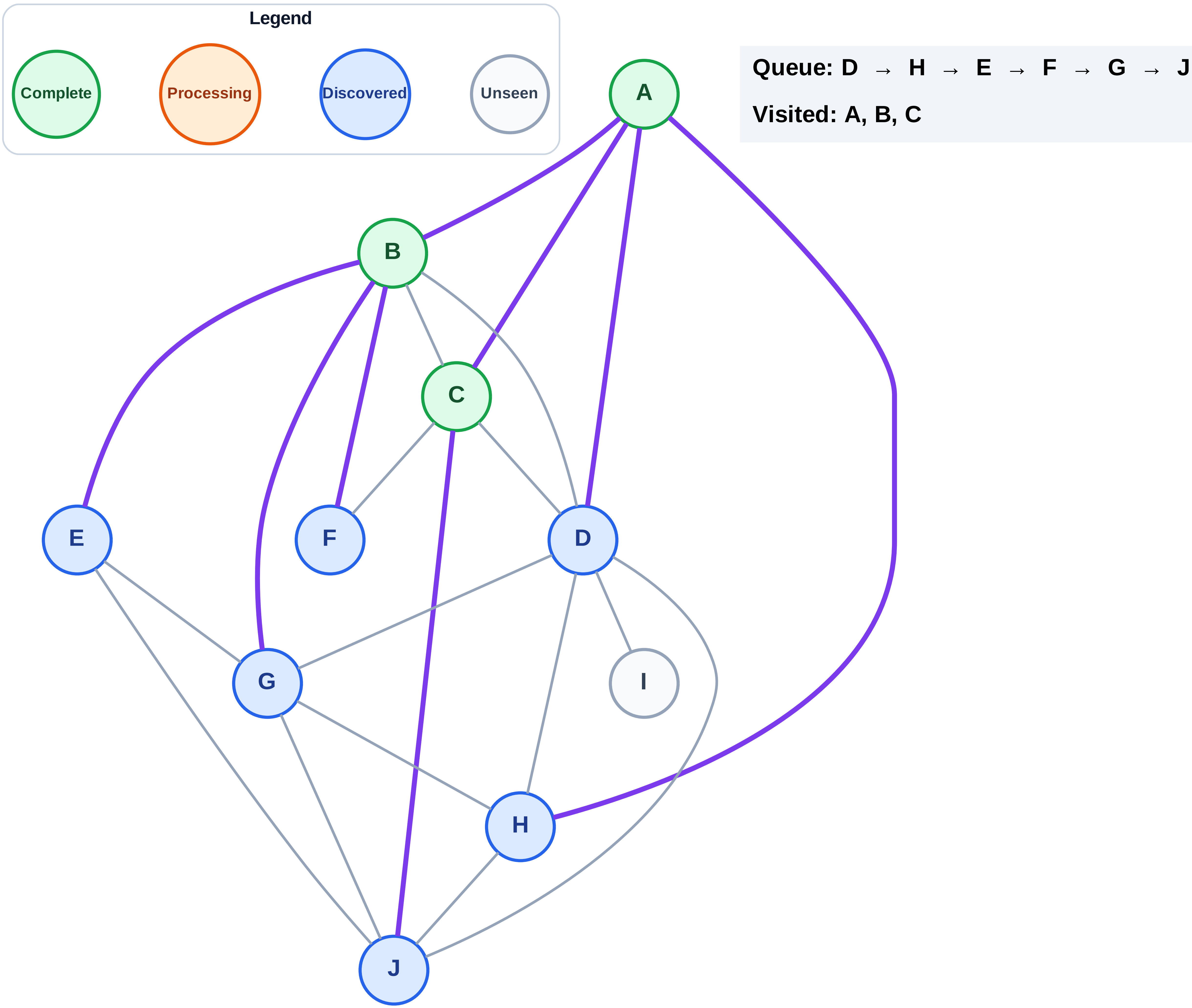
BFS Traversal

Step 6: Process node C



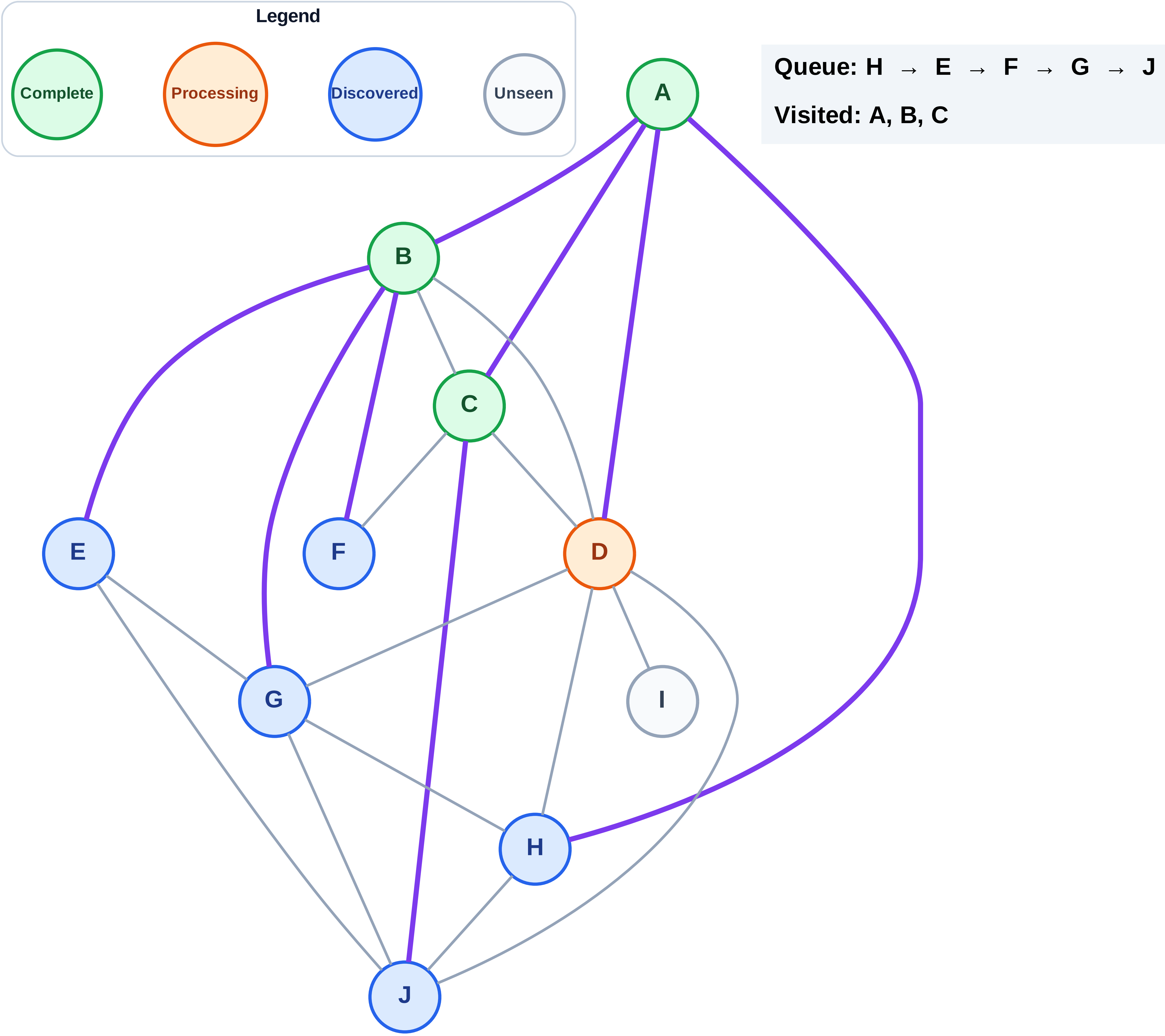
BFS Traversal

Step 7: Discover J from C



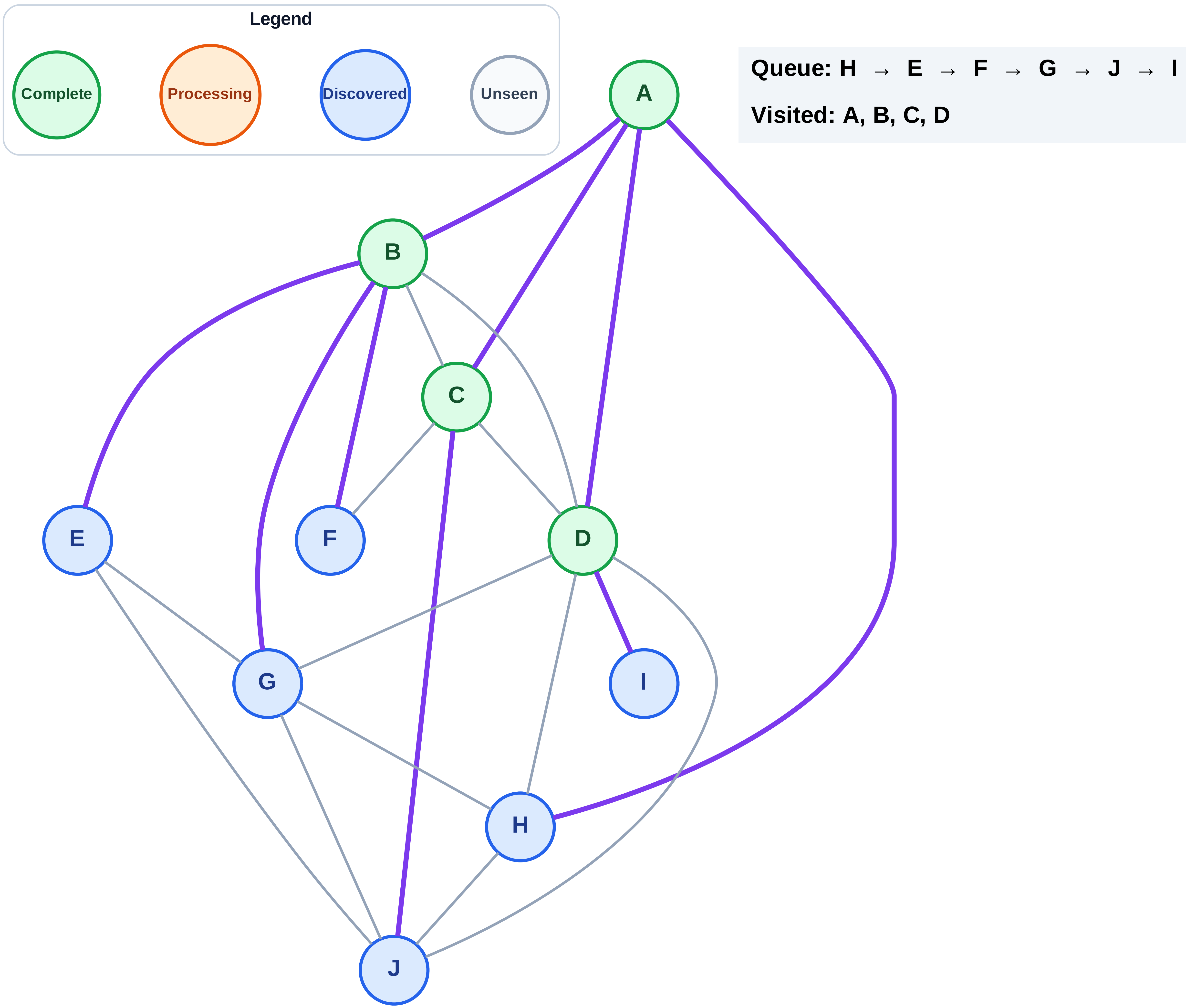
BFS Traversal

Step 8: Process node D



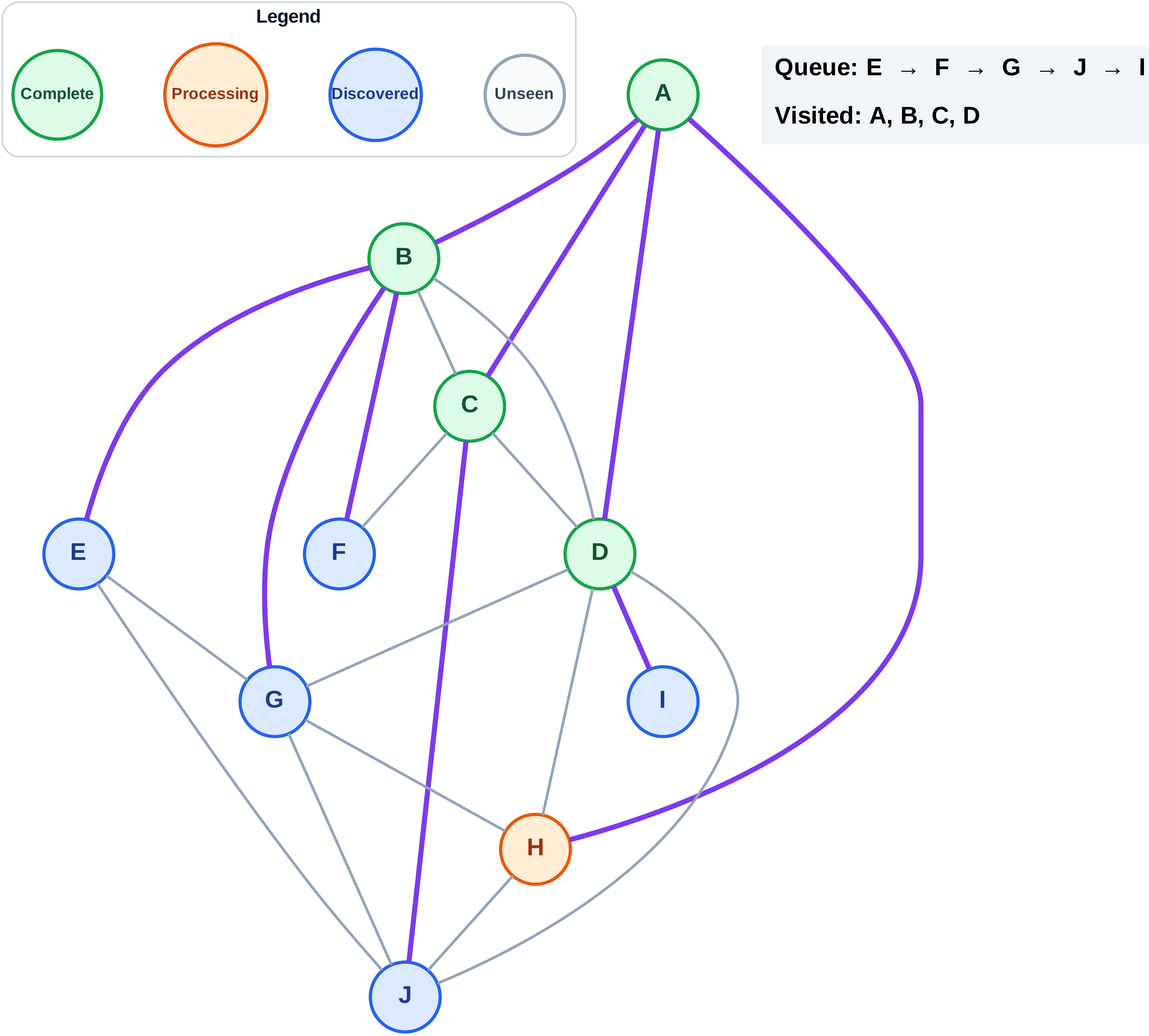
BFS Traversal

Step 9: Discover I from D



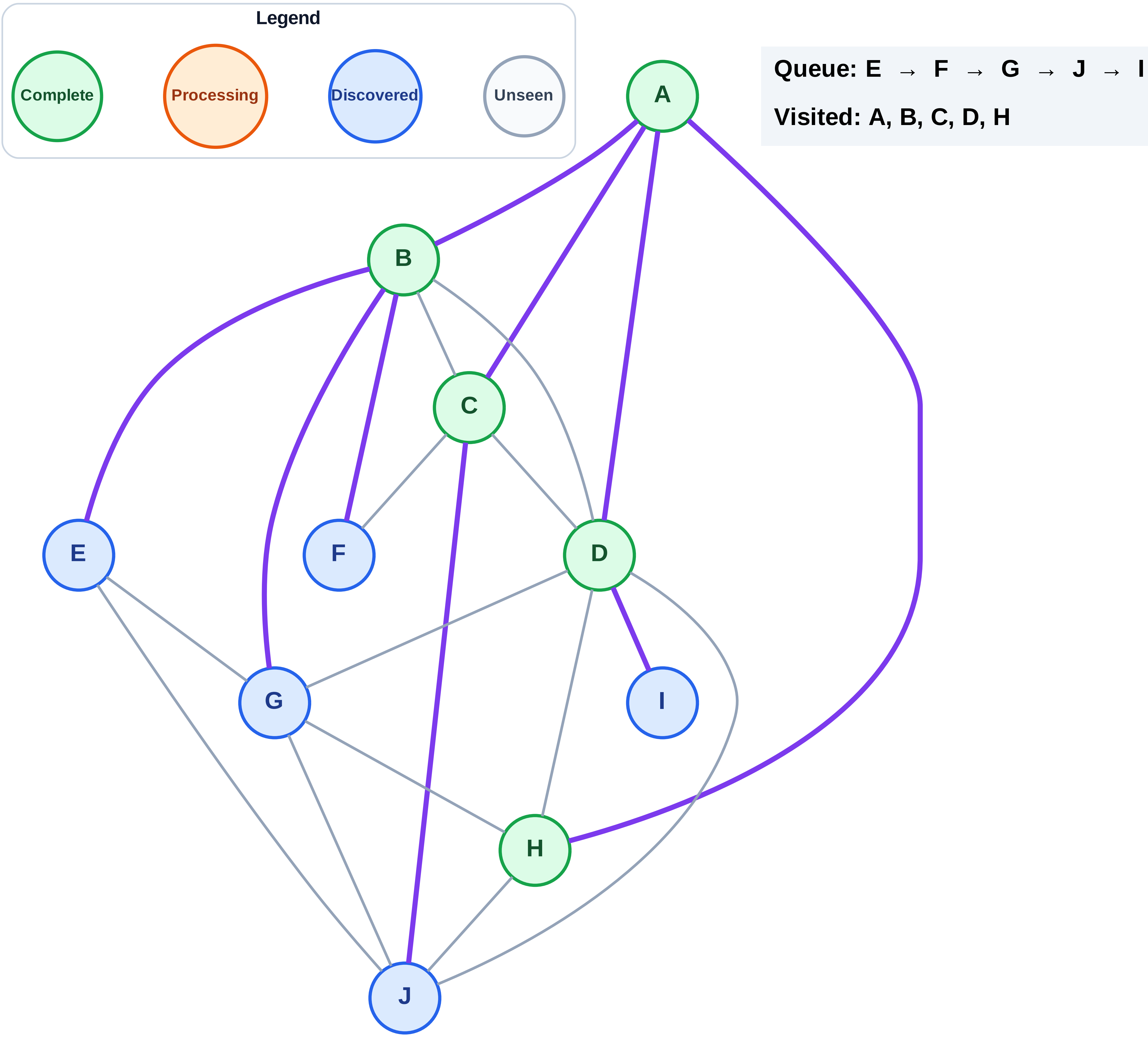
BFS Traversal

Step 10: Process node H



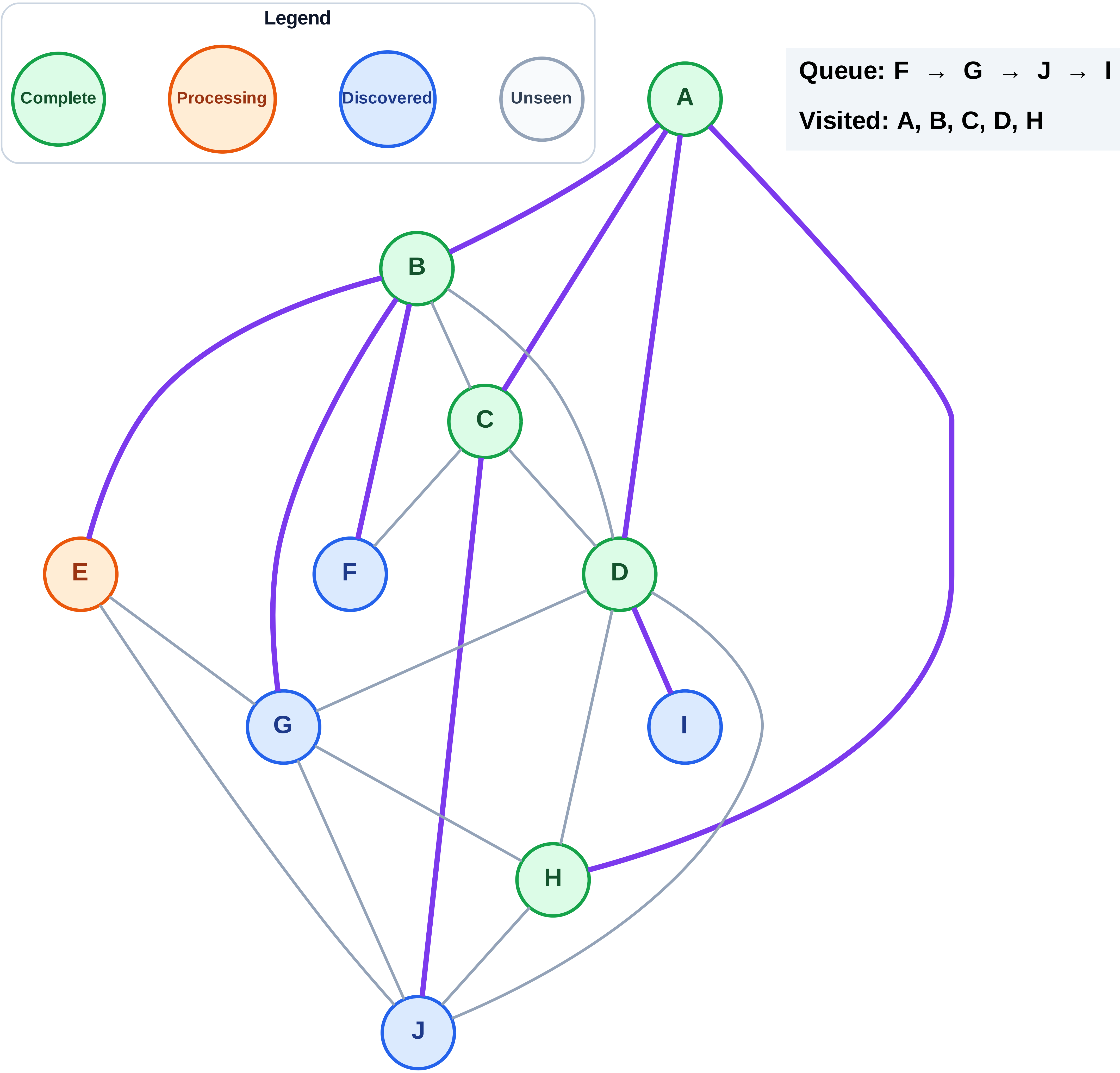
BFS Traversal

Step 11: No new neighbors from H



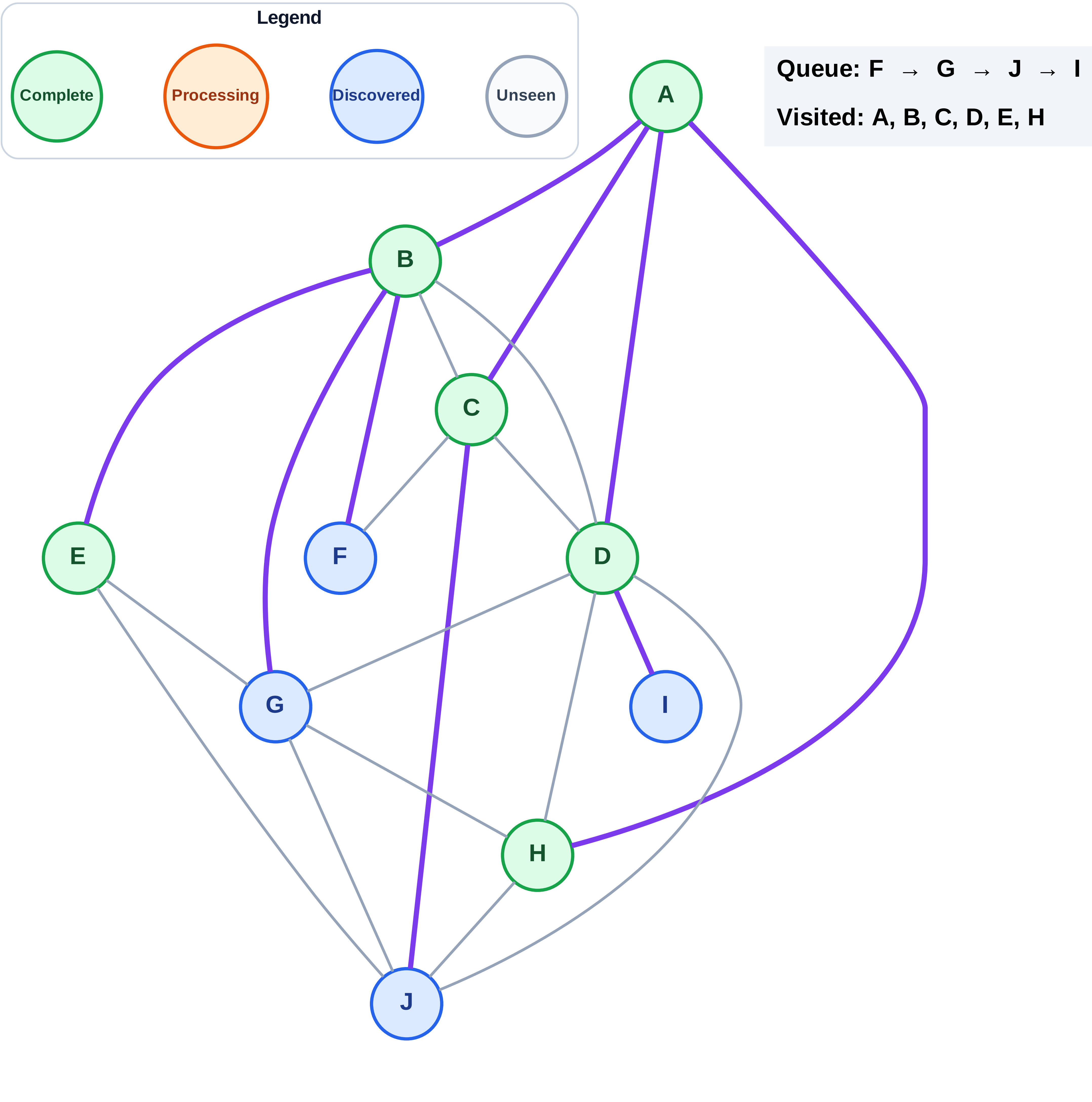
BFS Traversal

Step 12: Process node E



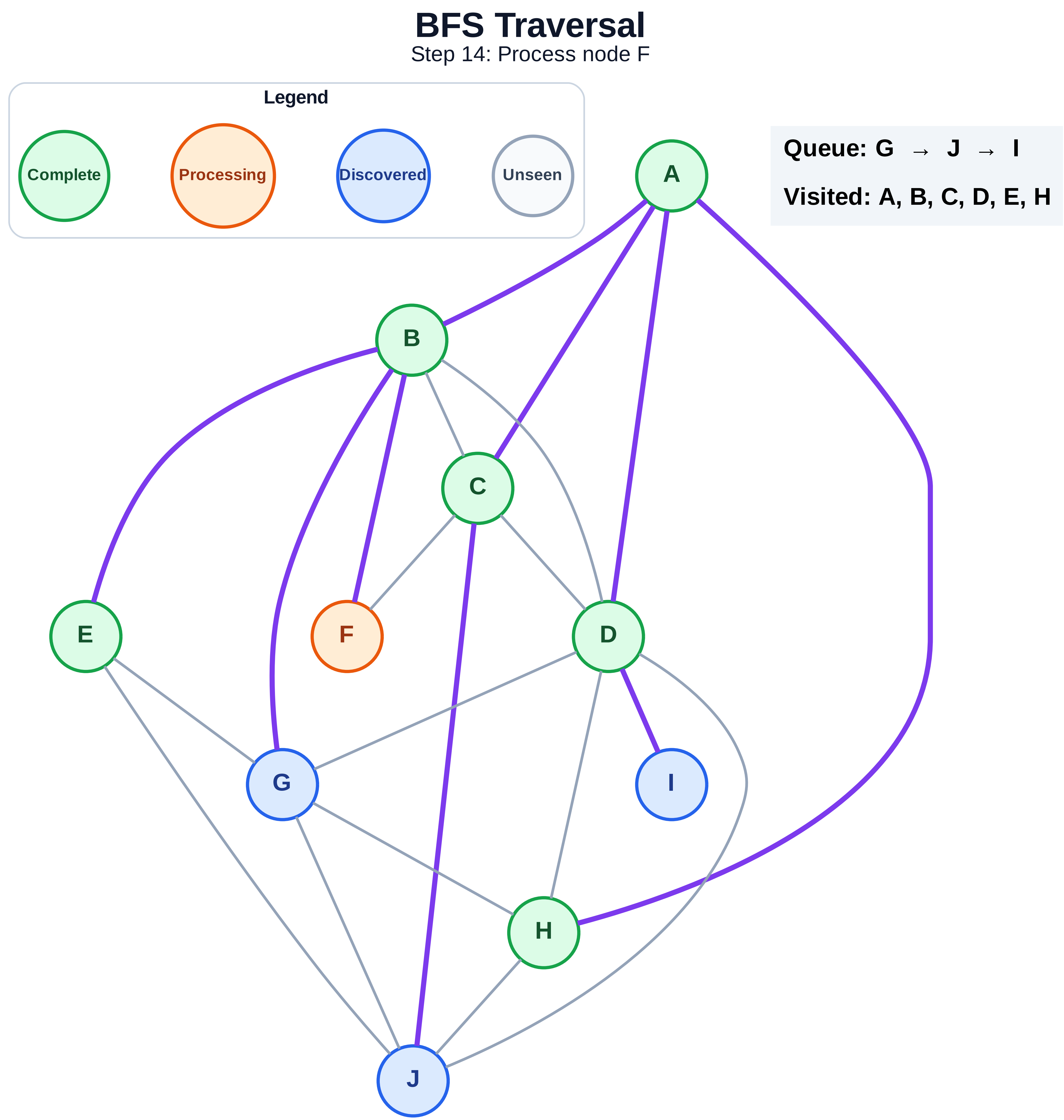
BFS Traversal

Step 13: No new neighbors from E



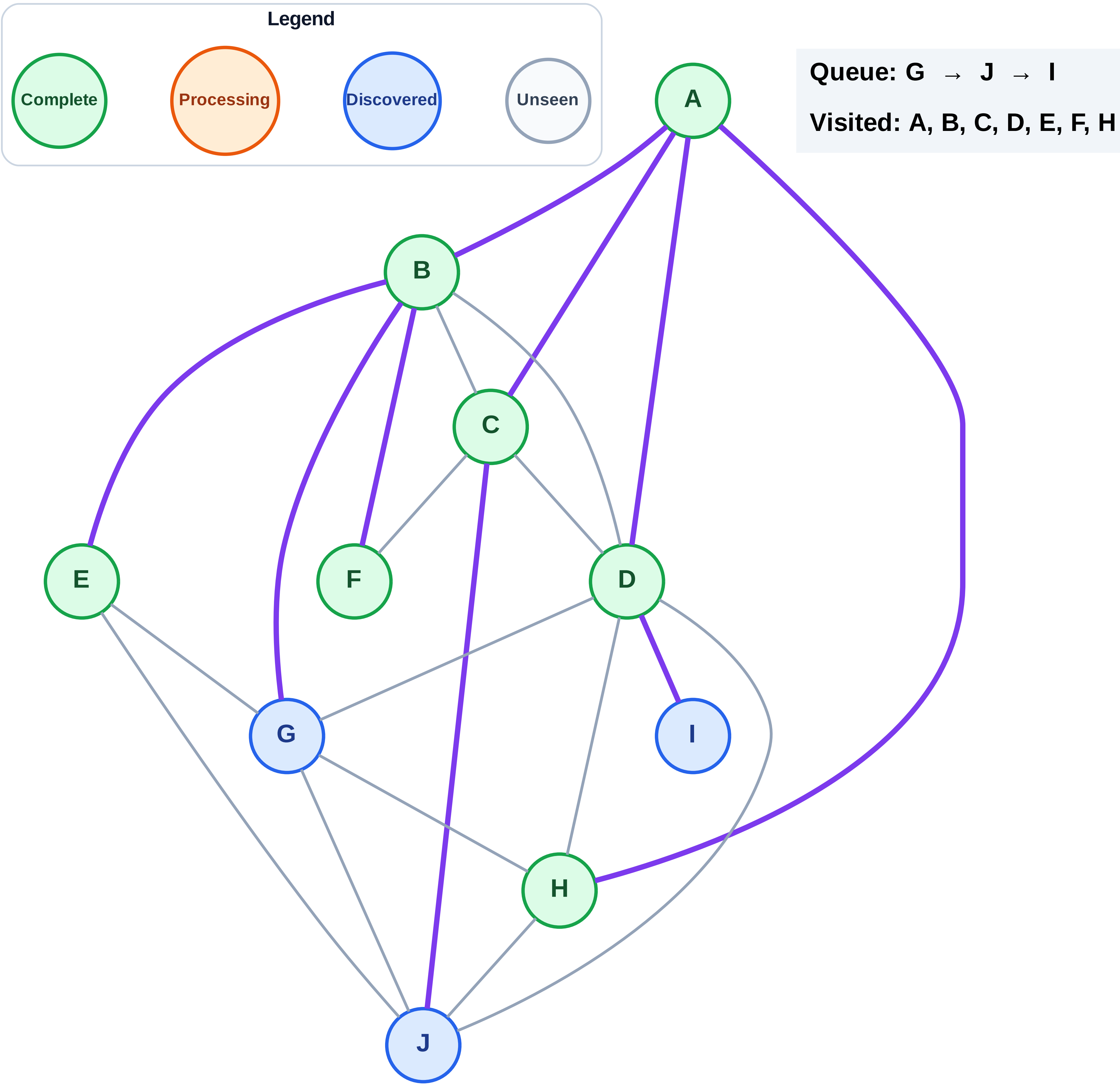
Queue: F → G → J → I

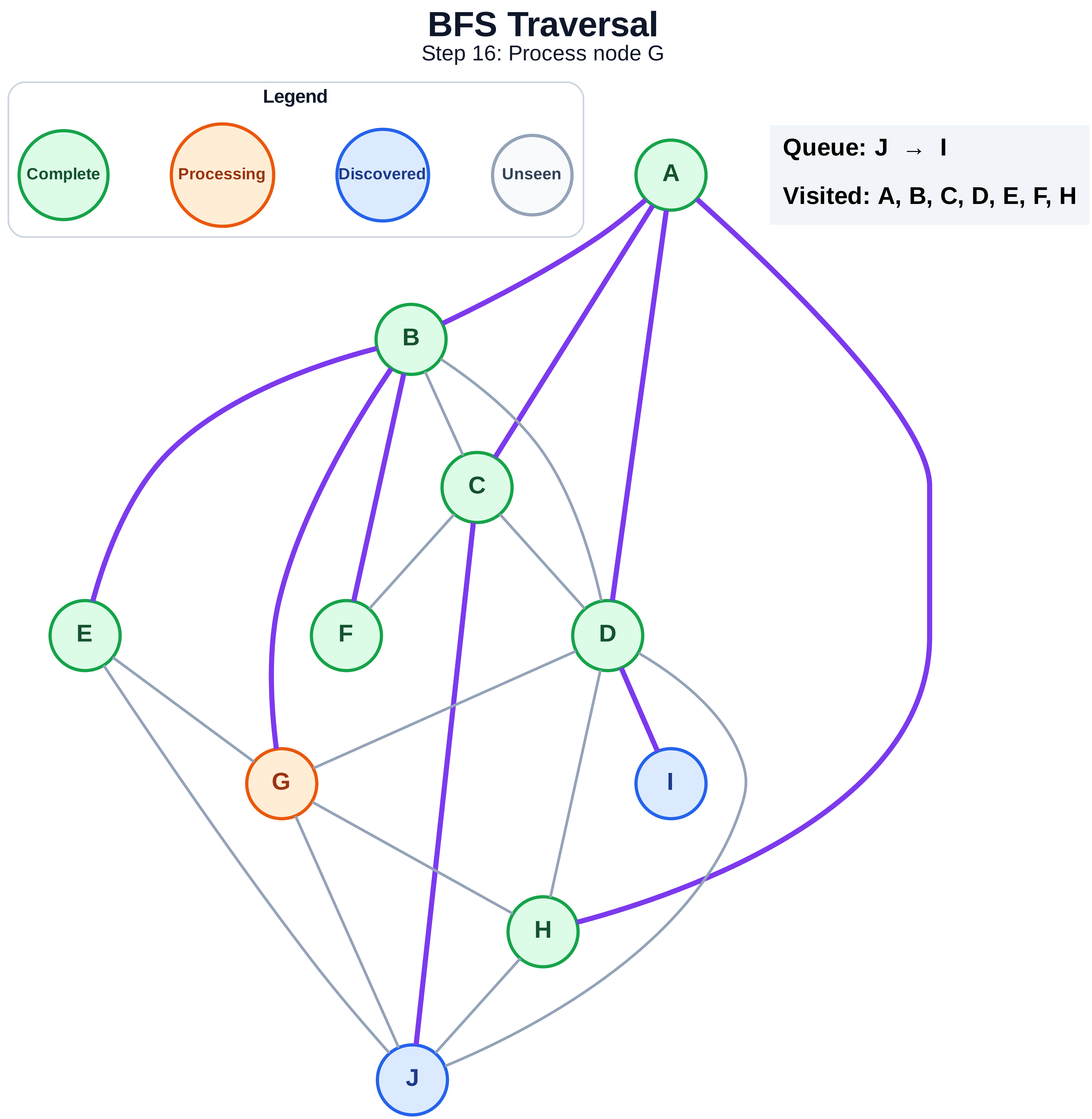
Visited: A, B, C, D, E, H


BFS Traversal
Step 14: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: G → J → I
Visited: A, B, C, D, E, H
A
B
C
D
E
F
G
H
I
J

BFS Traversal

Step 15: No new neighbors from F

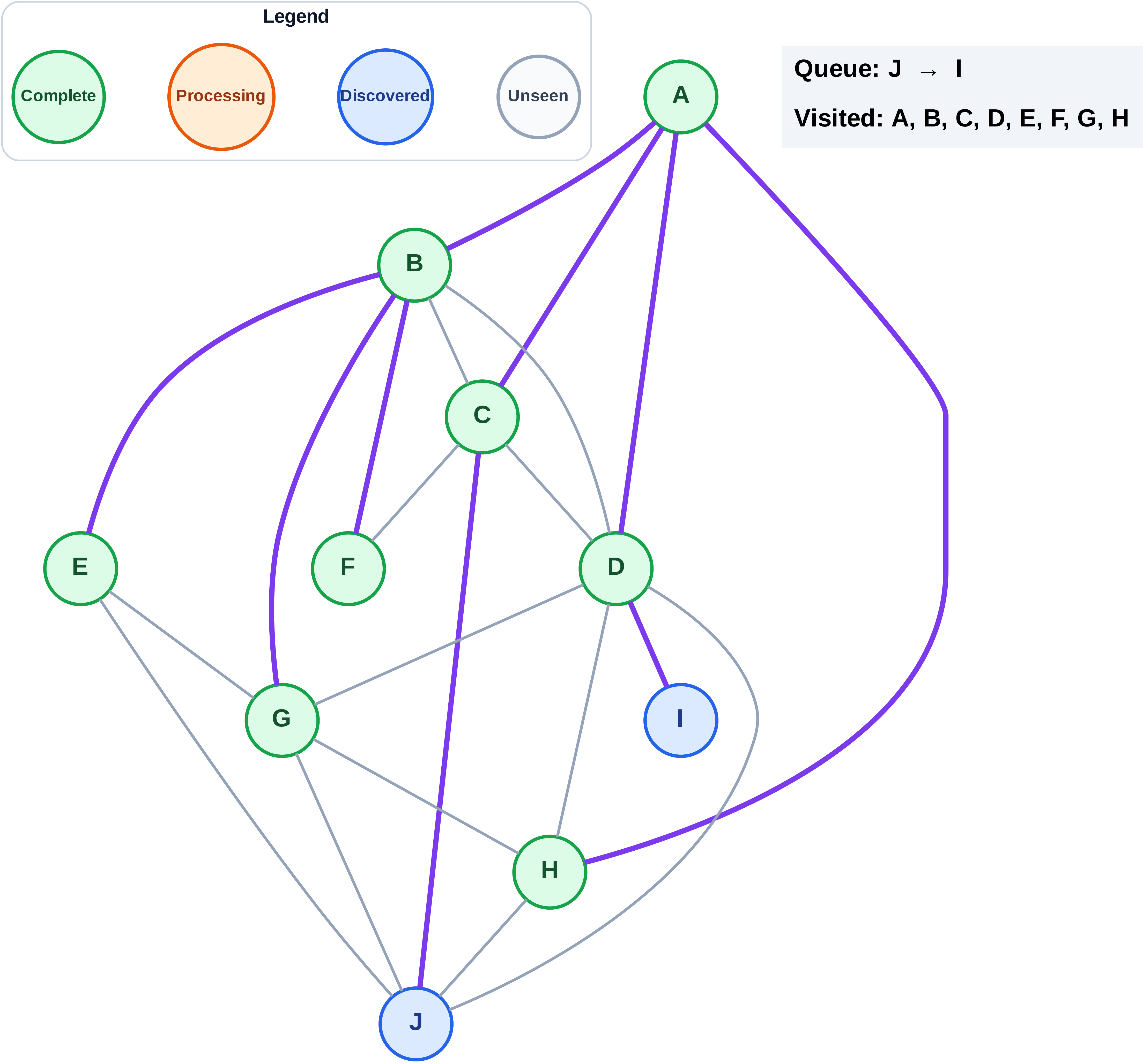


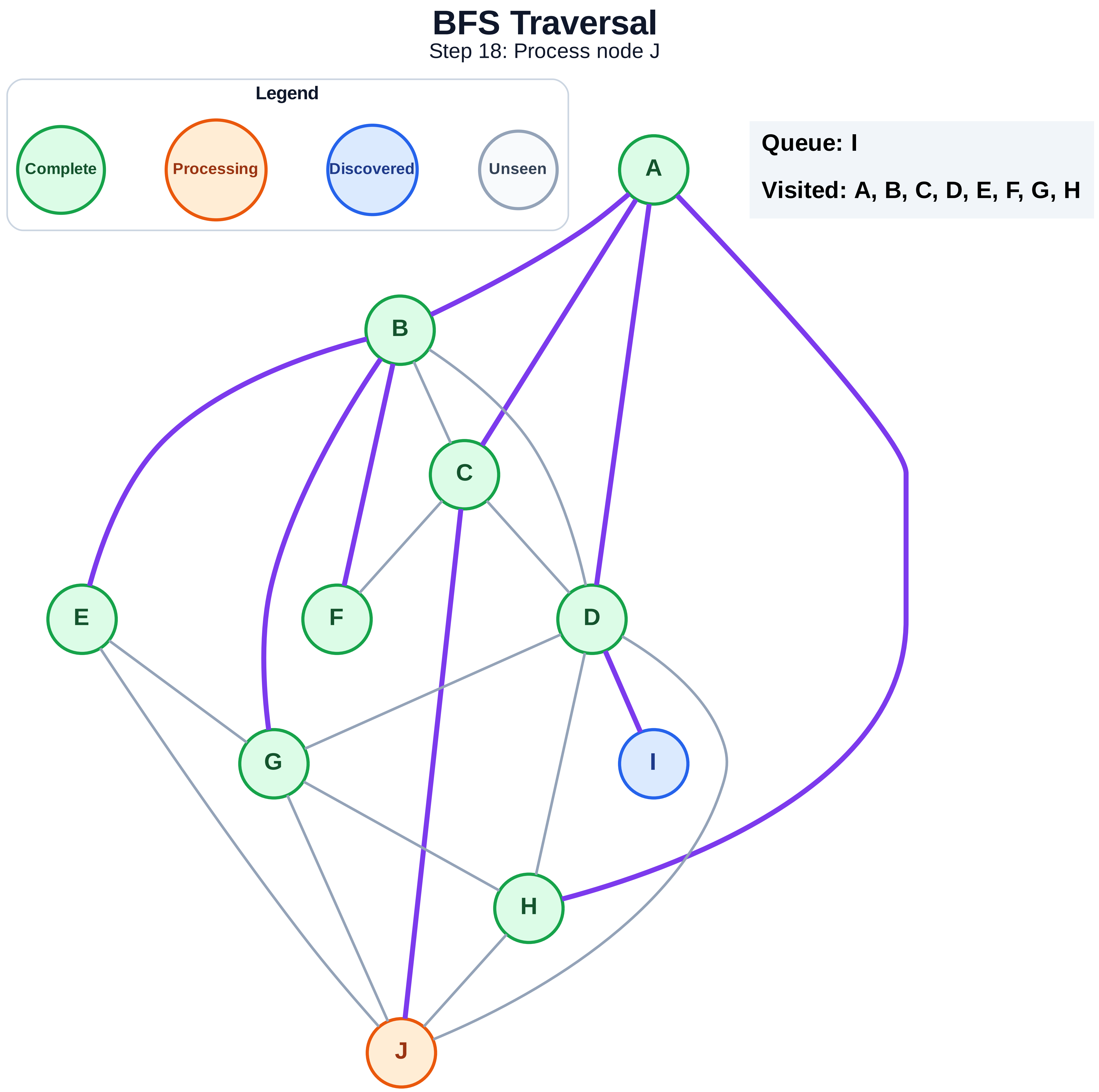


BFS Traversal
Step 16: Process node G
Legend
Complete
Processing
Discovered
Unseen
Queue: J → I
Visited: A, B, C, D, E, F, H
A
B
C
D
E
F
G
H
I
J

BFS Traversal

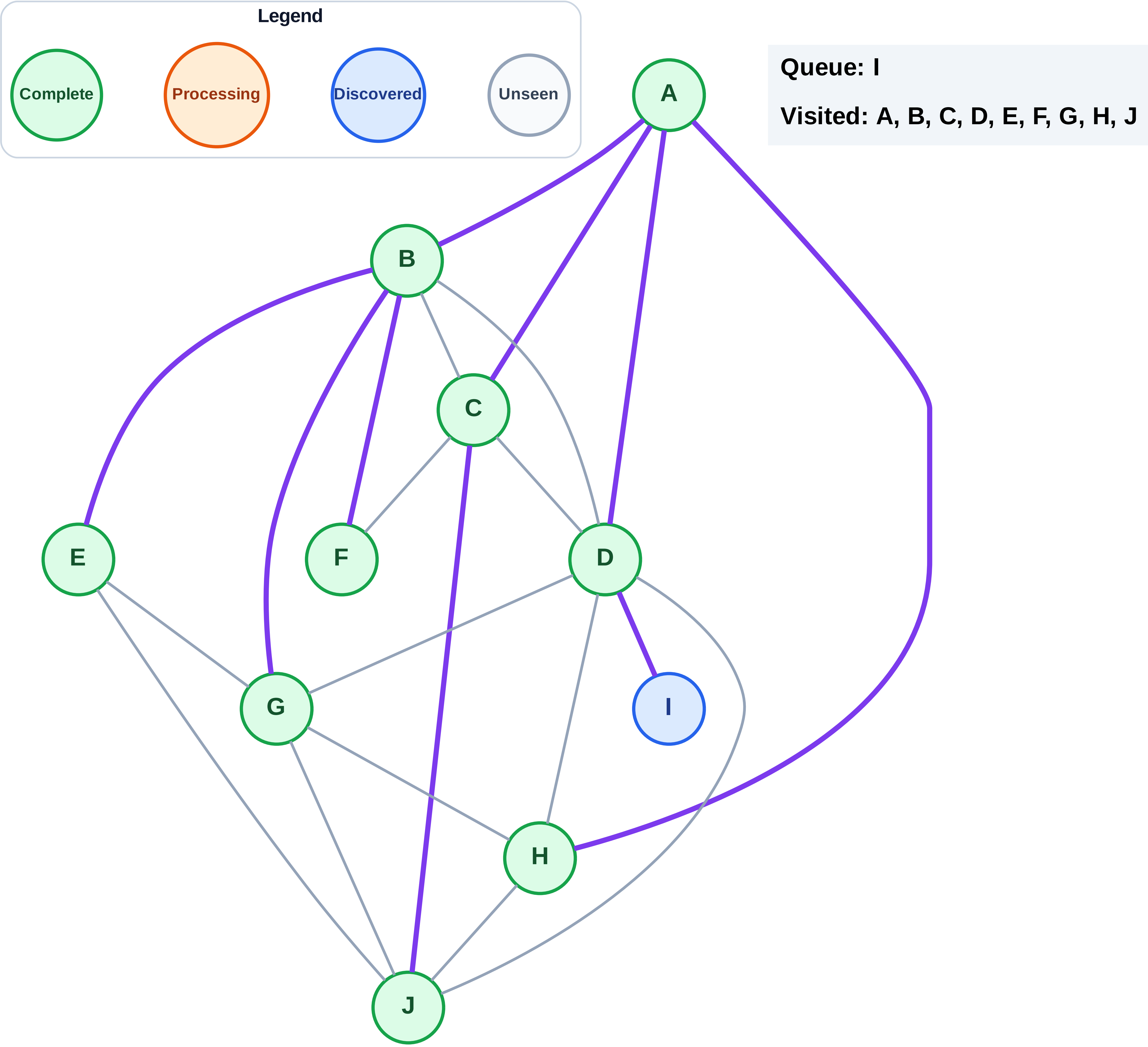
Step 17: No new neighbors from G




BFS Traversal
Step 18: Process node J
Legend
Complete
Processing
Discovered
Unseen
Queue: I
Visited: A, B, C, D, E, F, G, H
A
B
C
D
E
F
G
H
I
J

BFS Traversal

Step 19: No new neighbors from J

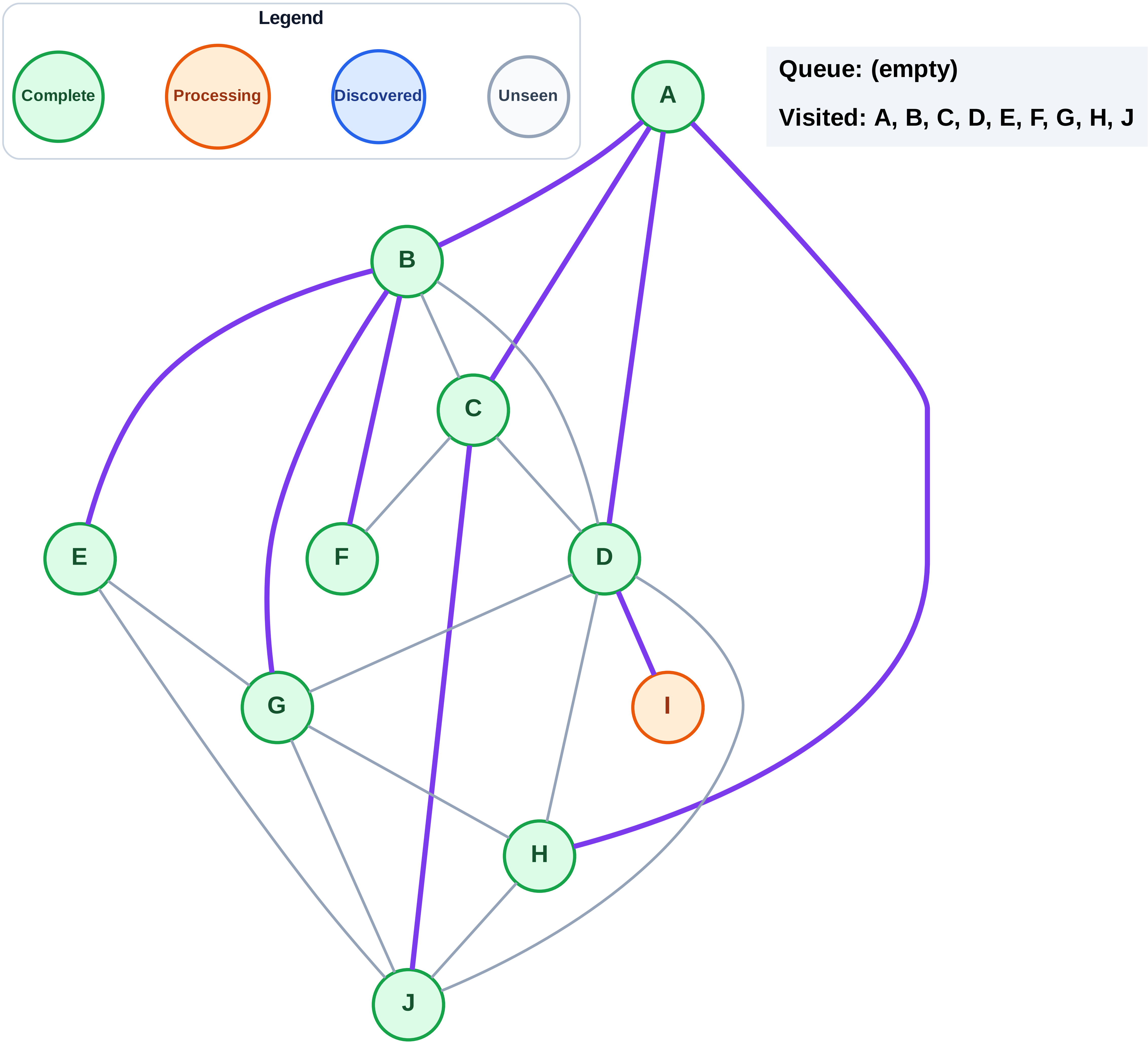


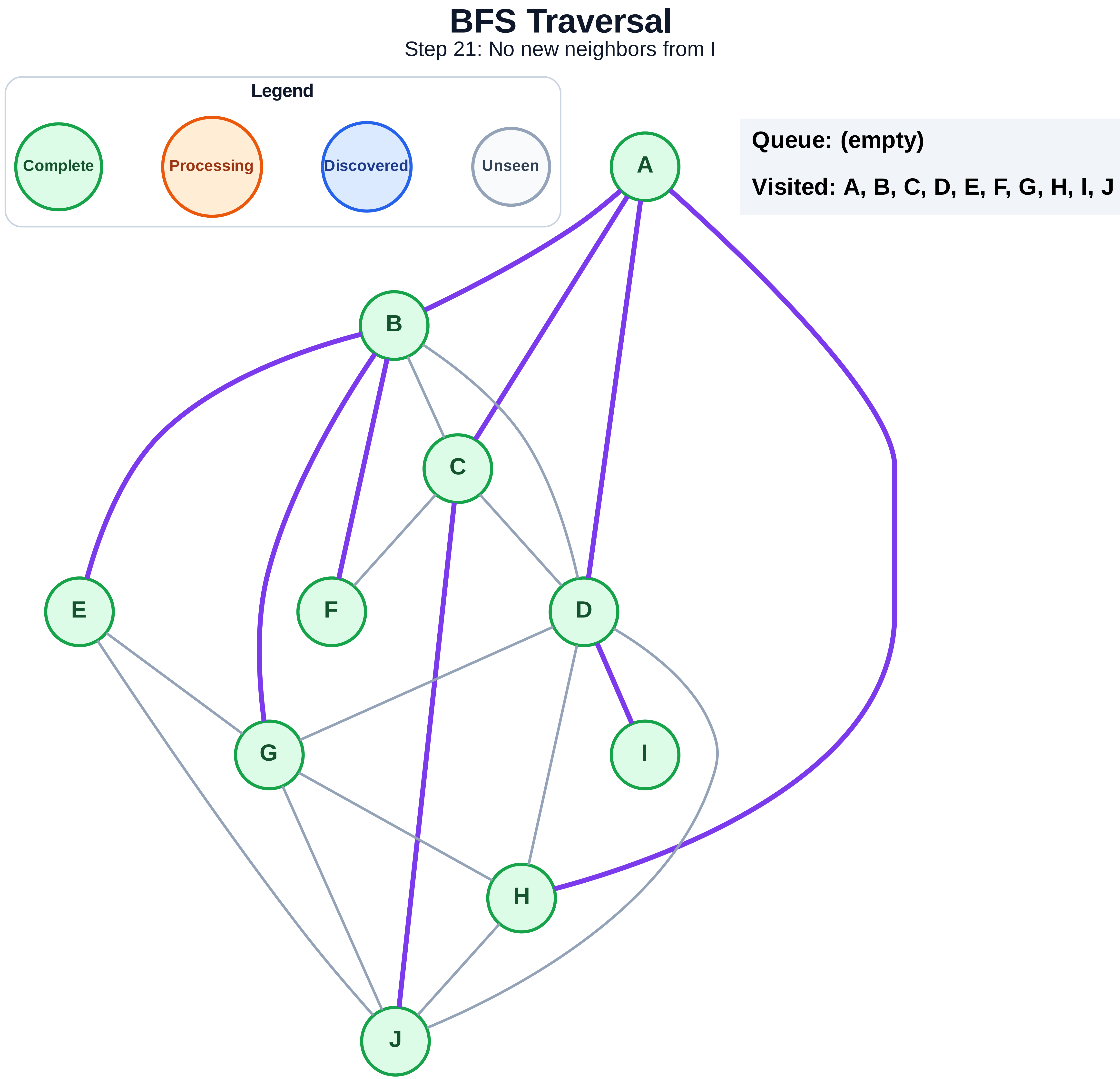
Queue: I

Visited: A, B, C, D, E, F, G, H, J

BFS Traversal

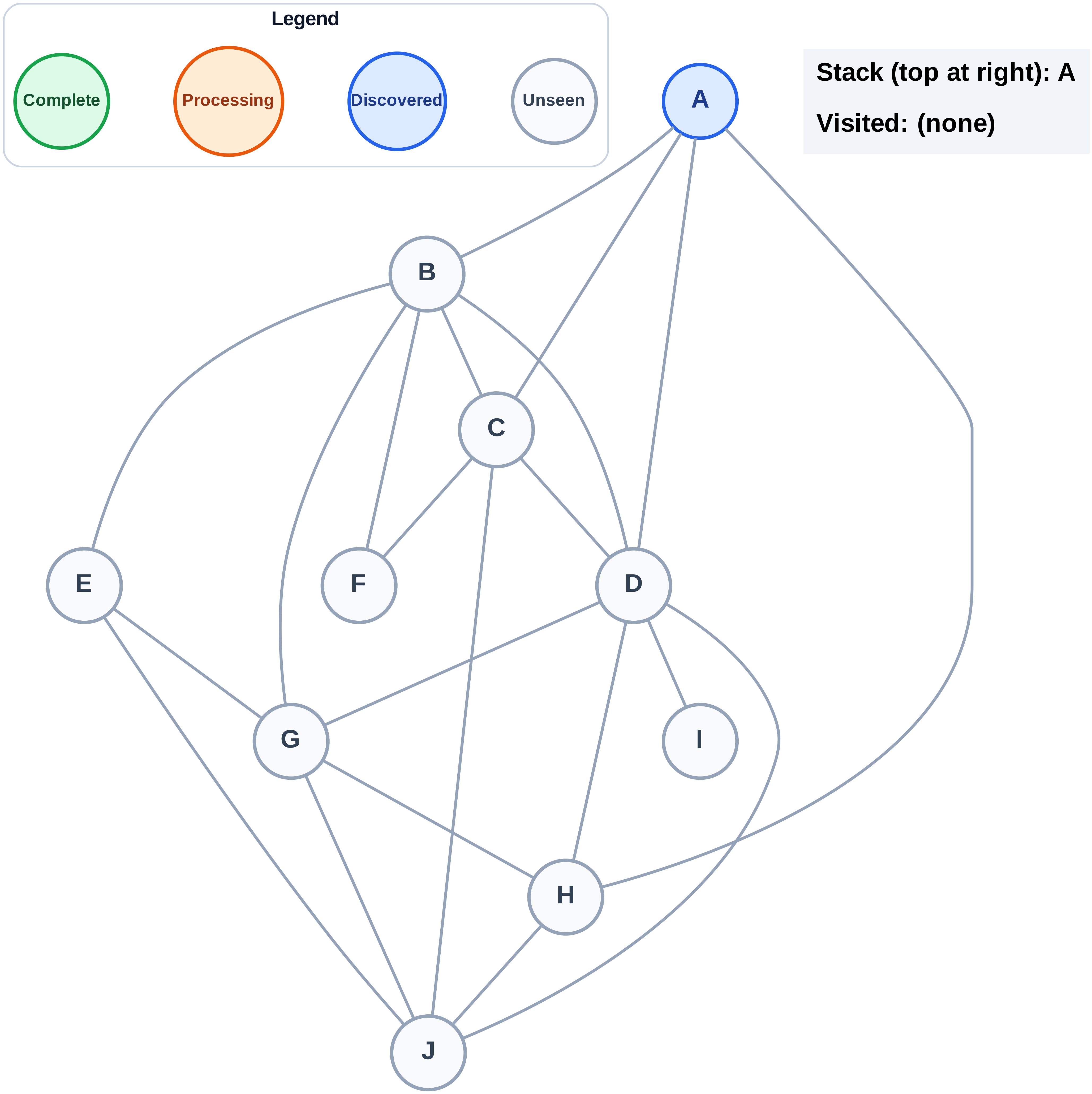
Step 20: Process node I




BFS Traversal
Step 21: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 1: Start at A

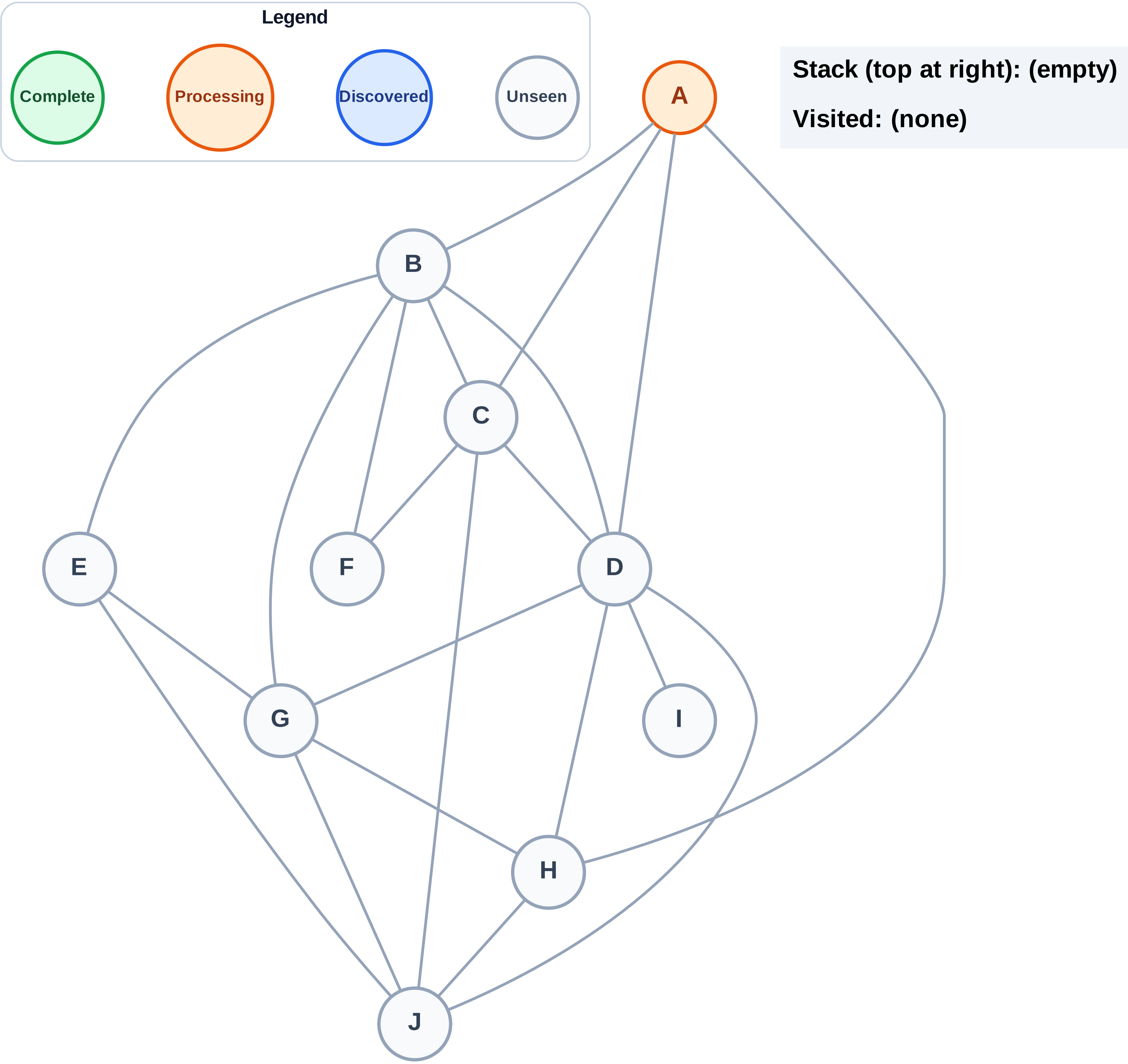


Stack (top at right): A

Visited: (none)

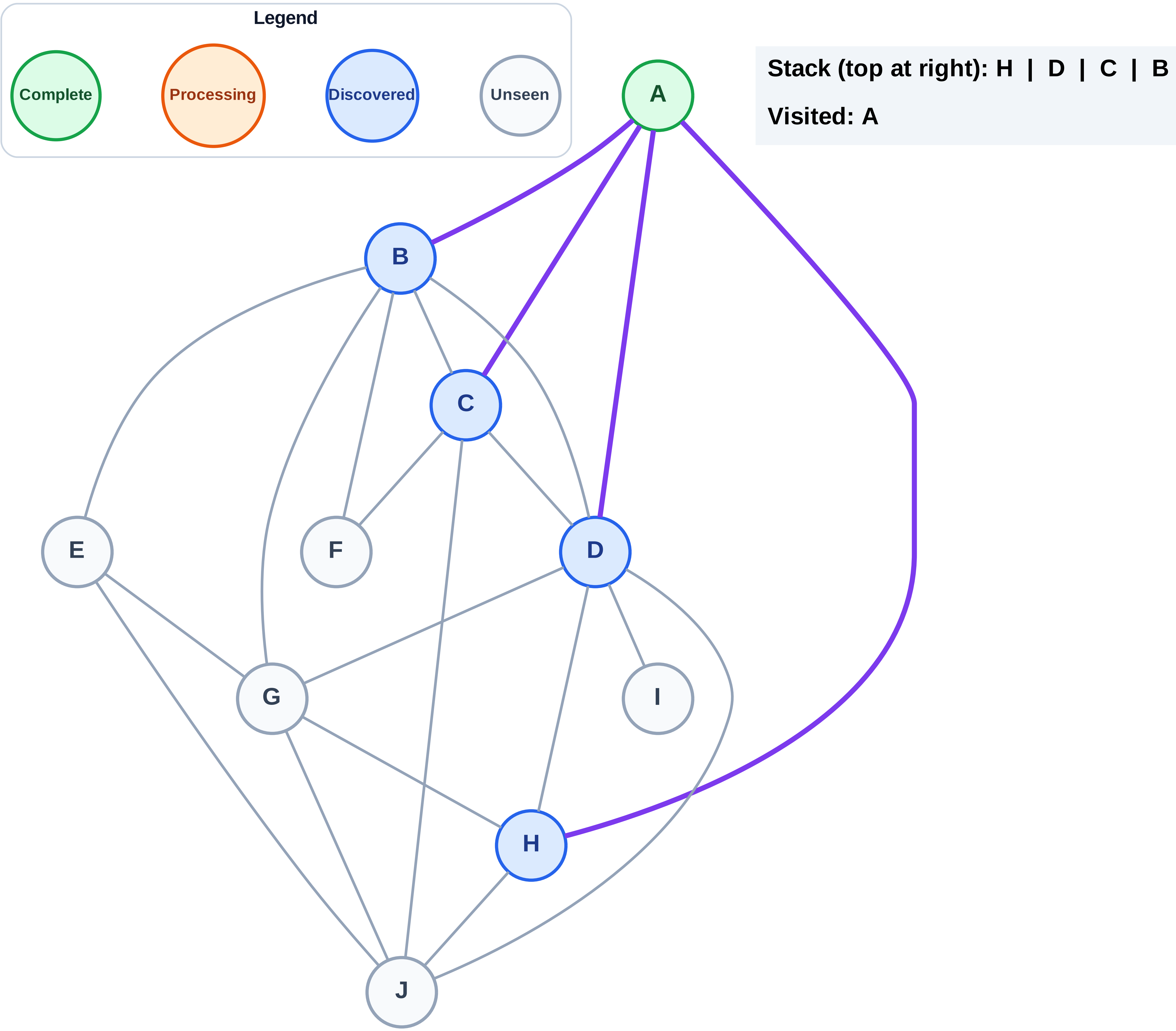
DFS Traversal

Step 2: Process node A



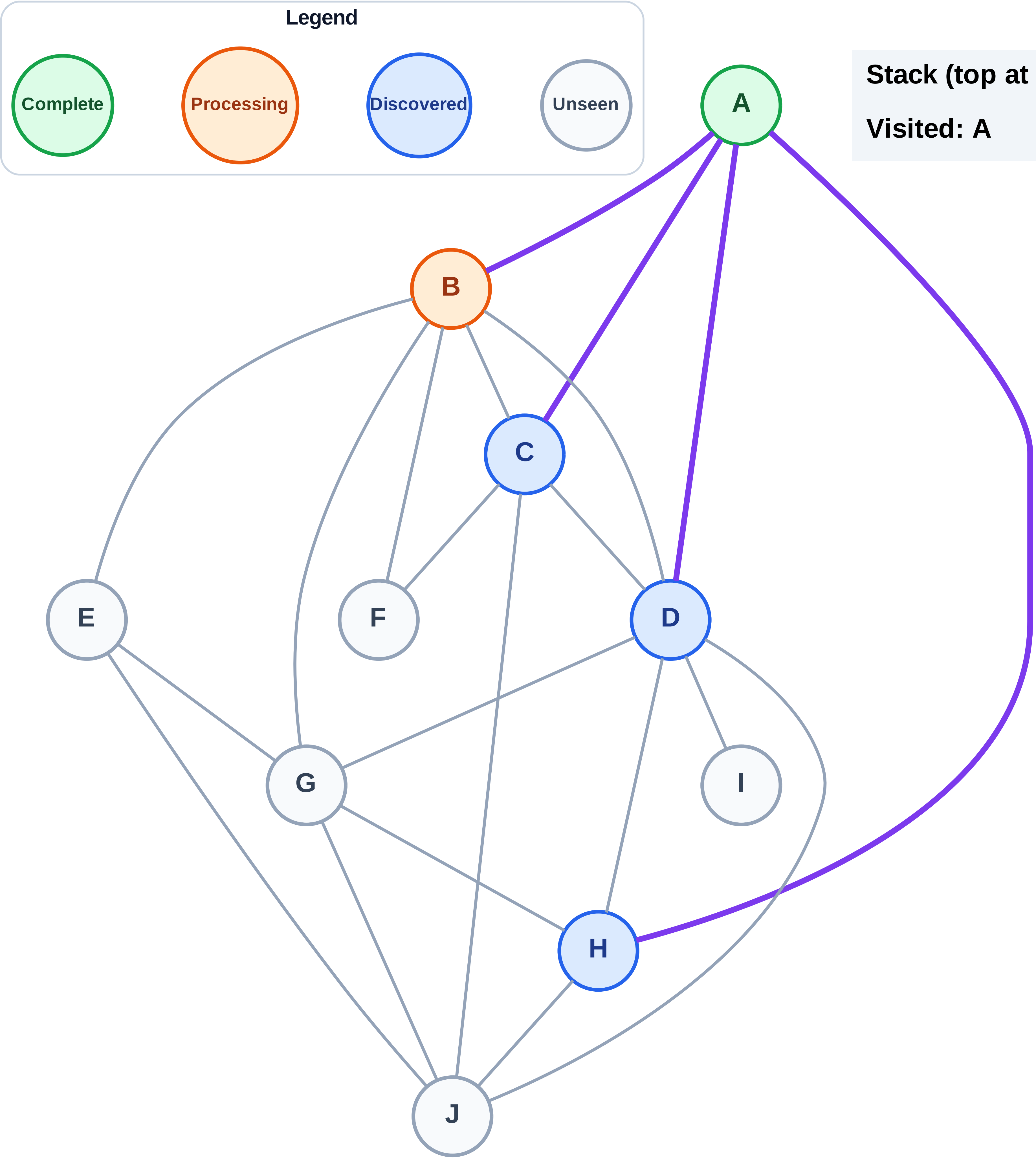
DFS Traversal

Step 3: Discover H, D, C, B from A



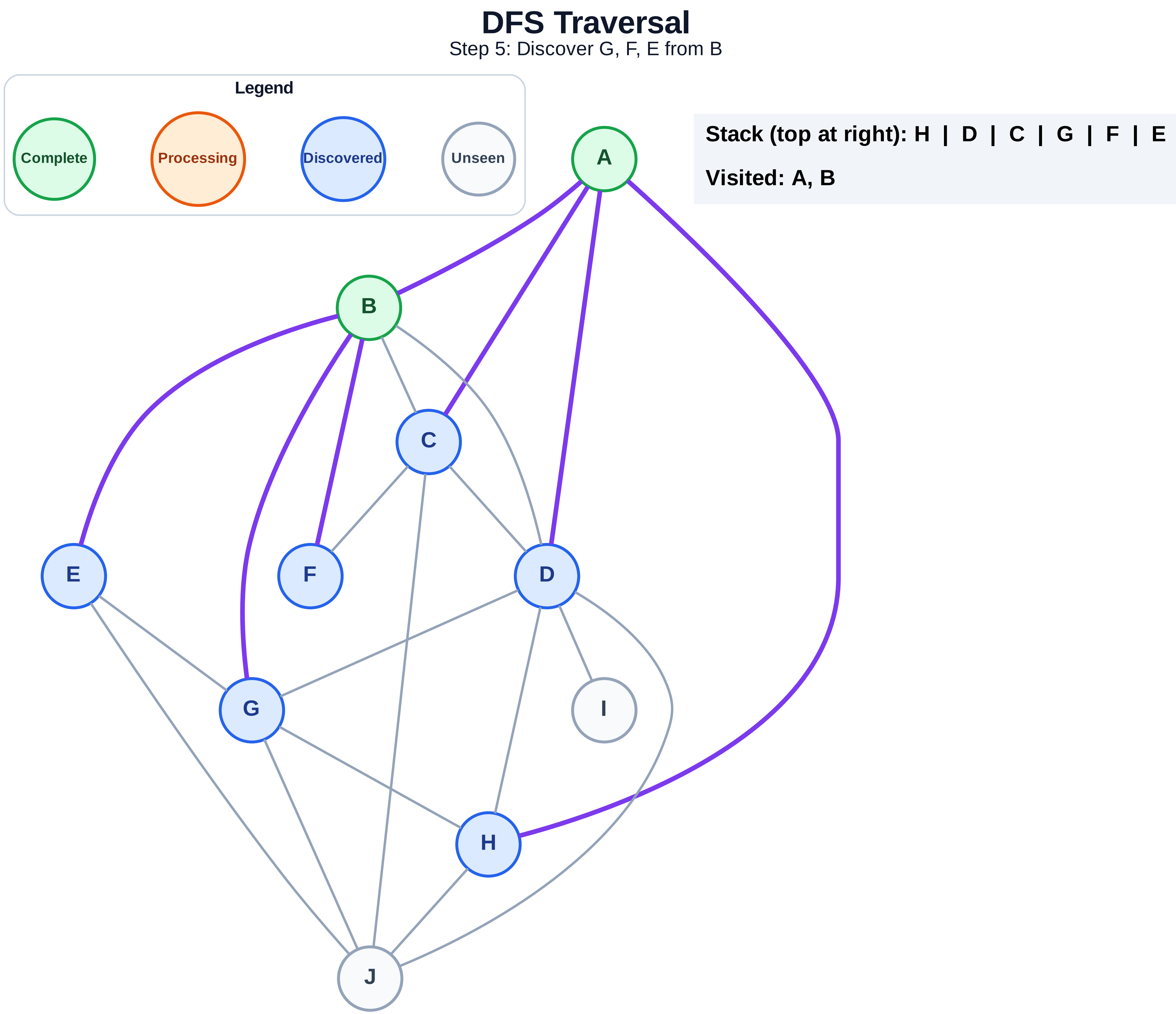
DFS Traversal

Step 4: Process node B



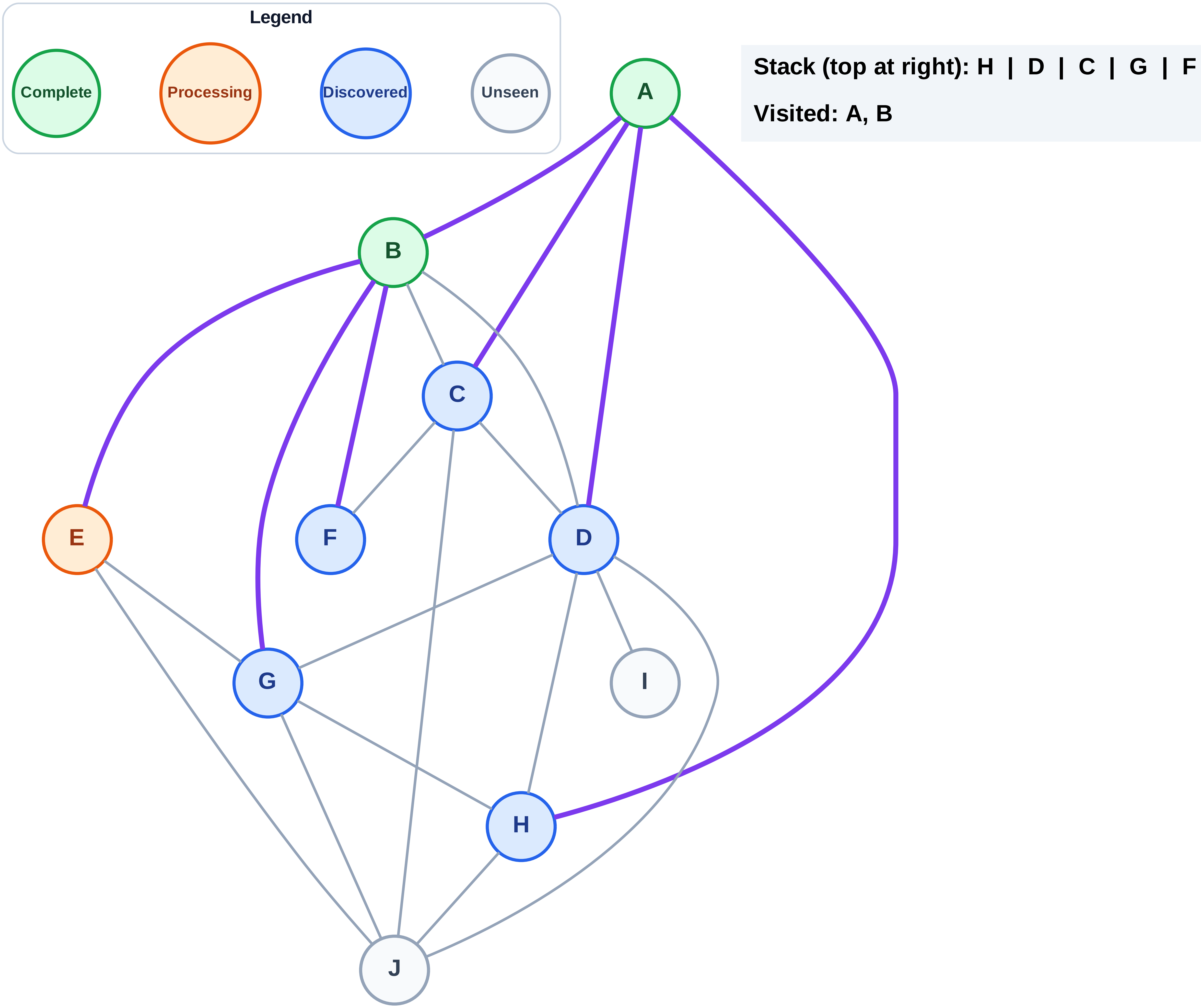
Stack (top at right): H | D | C

Visited: A


DFS Traversal
Step 5: Discover G, F, E from B
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | D | C | G | F | E
Visited: A, B
A
B
C
D
E
F
G
H
I
J

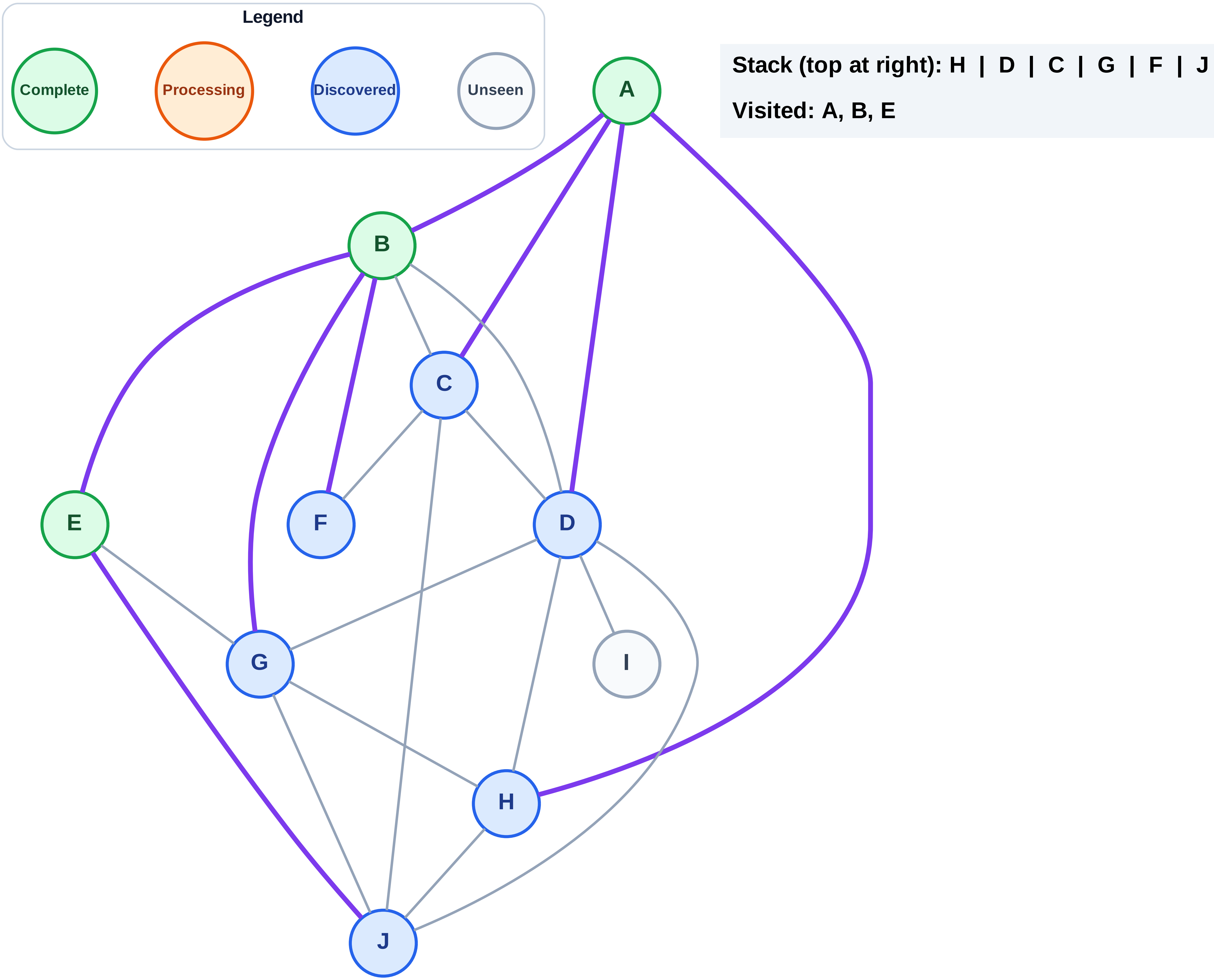
DFS Traversal

Step 6: Process node E



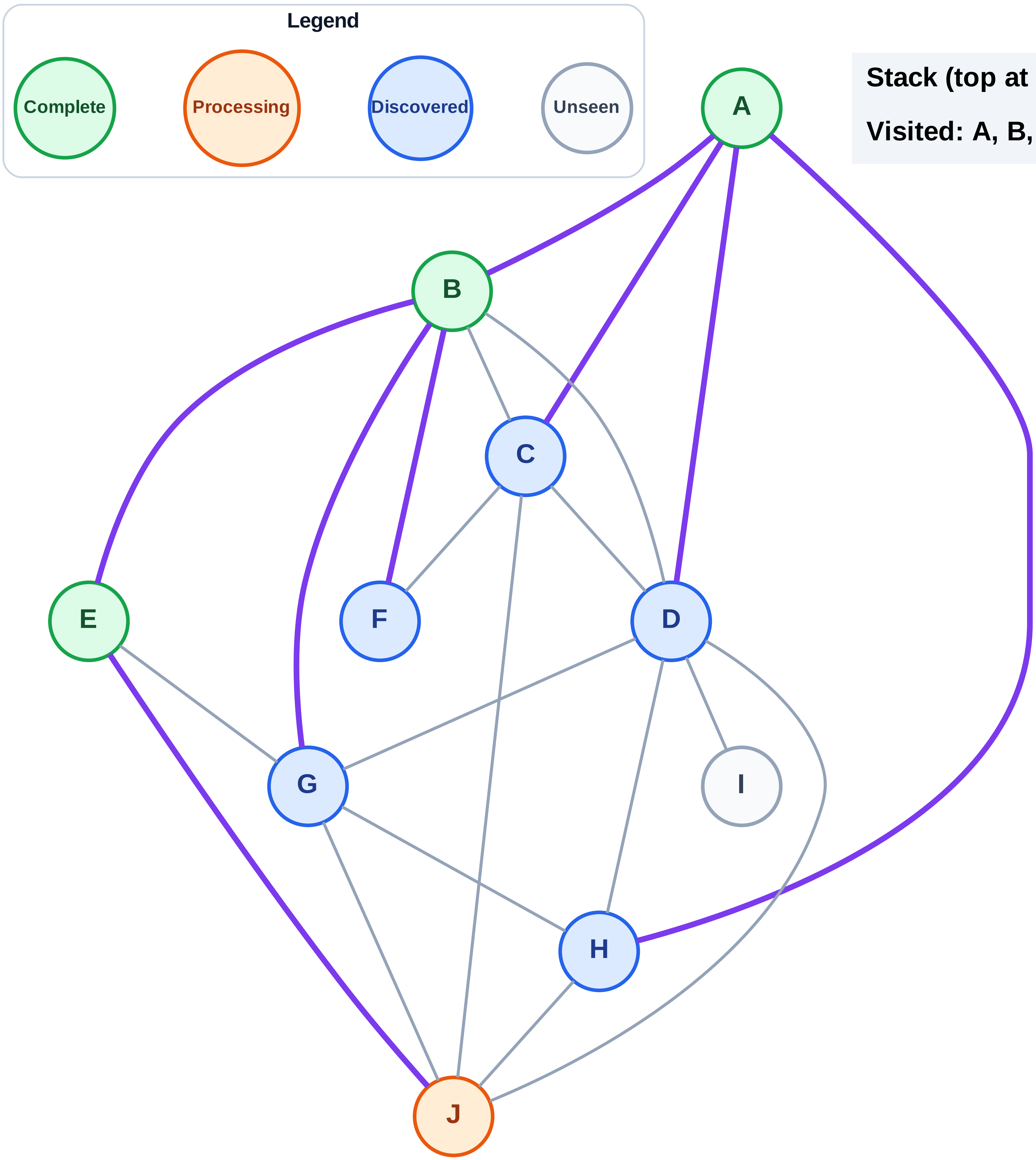
DFS Traversal

Step 7: Discover J from E



DFS Traversal

Step 8: Process node J

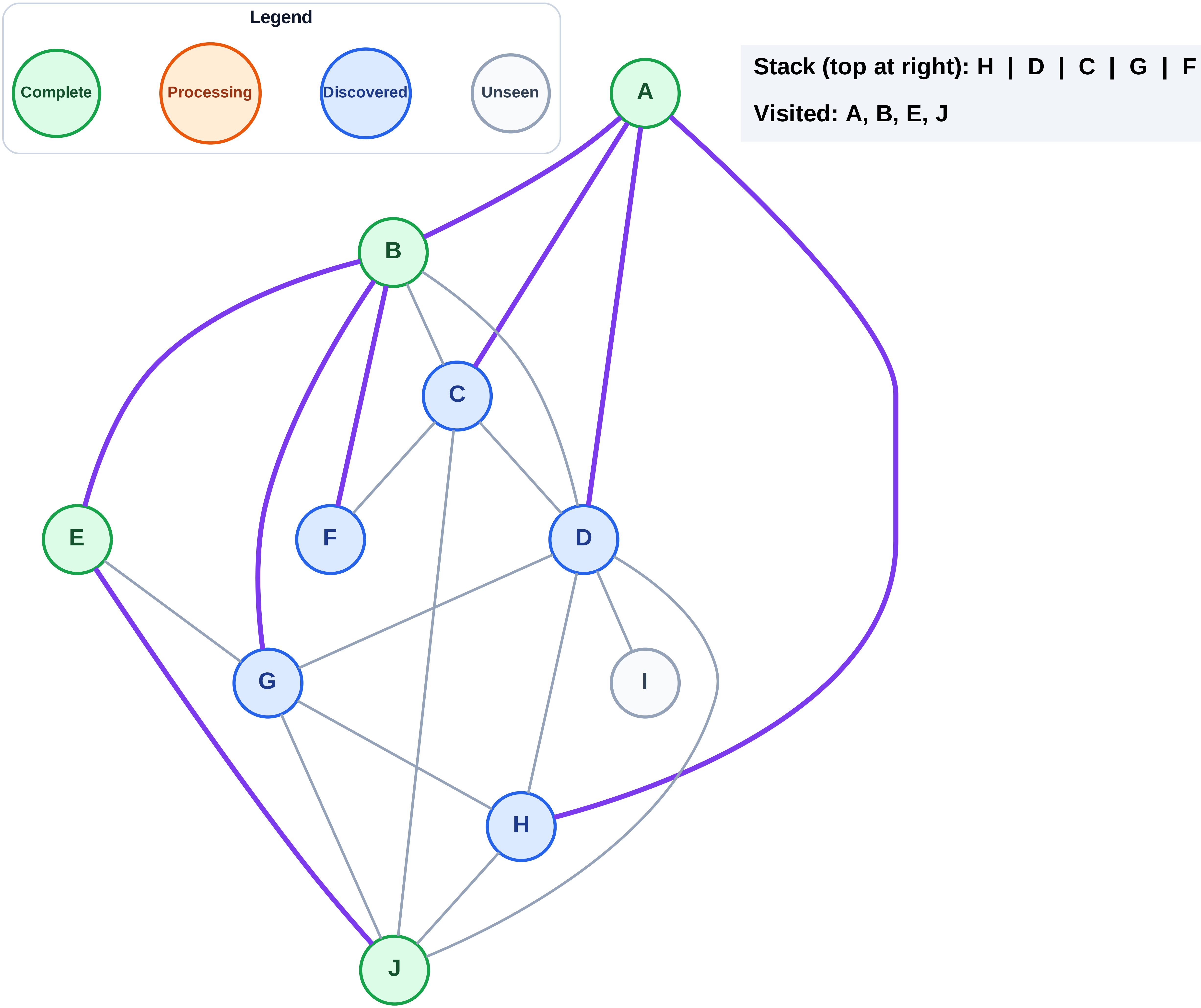


Stack (top at right): H | D | C | G | F

Visited: A, B, E

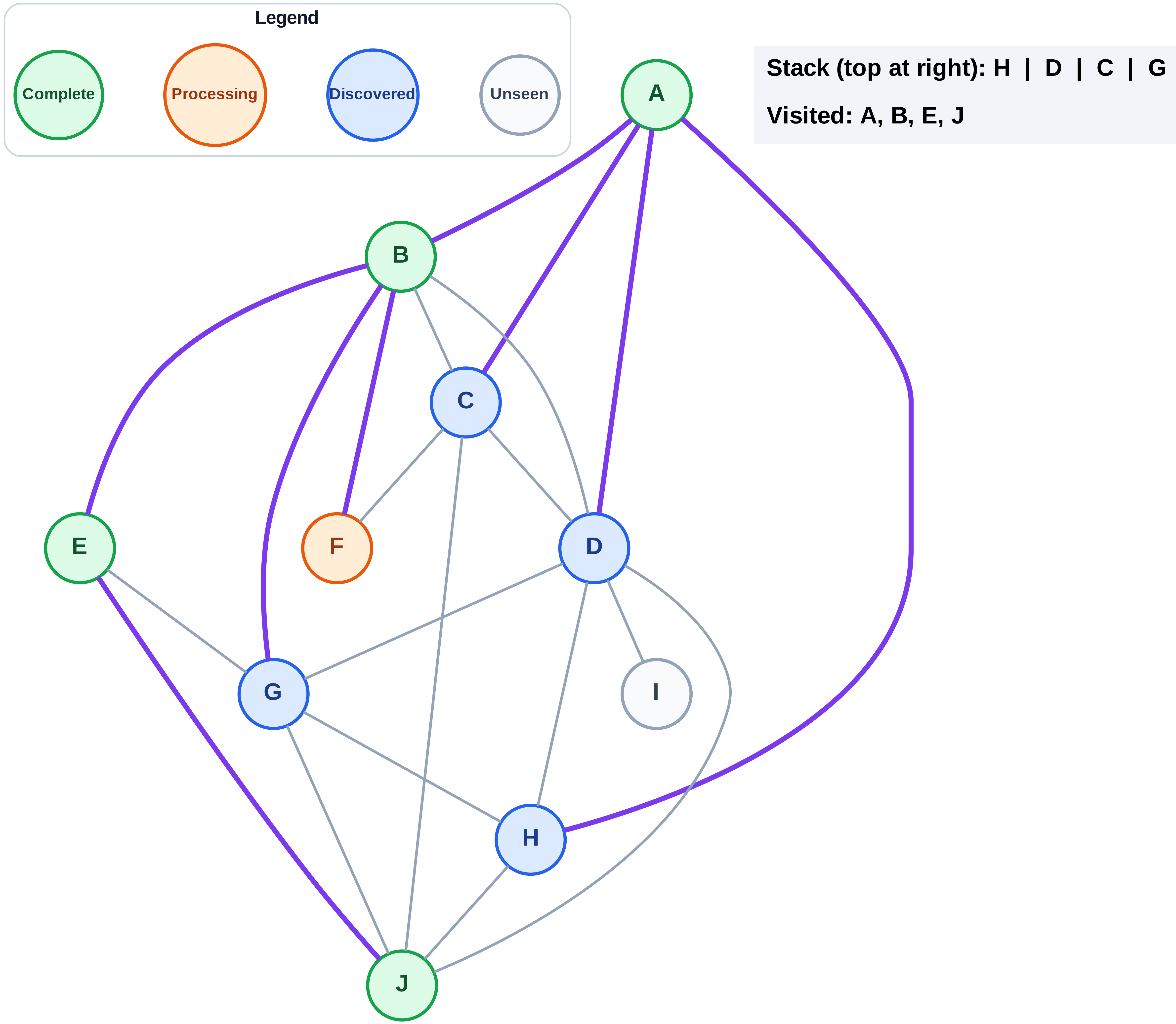
DFS Traversal

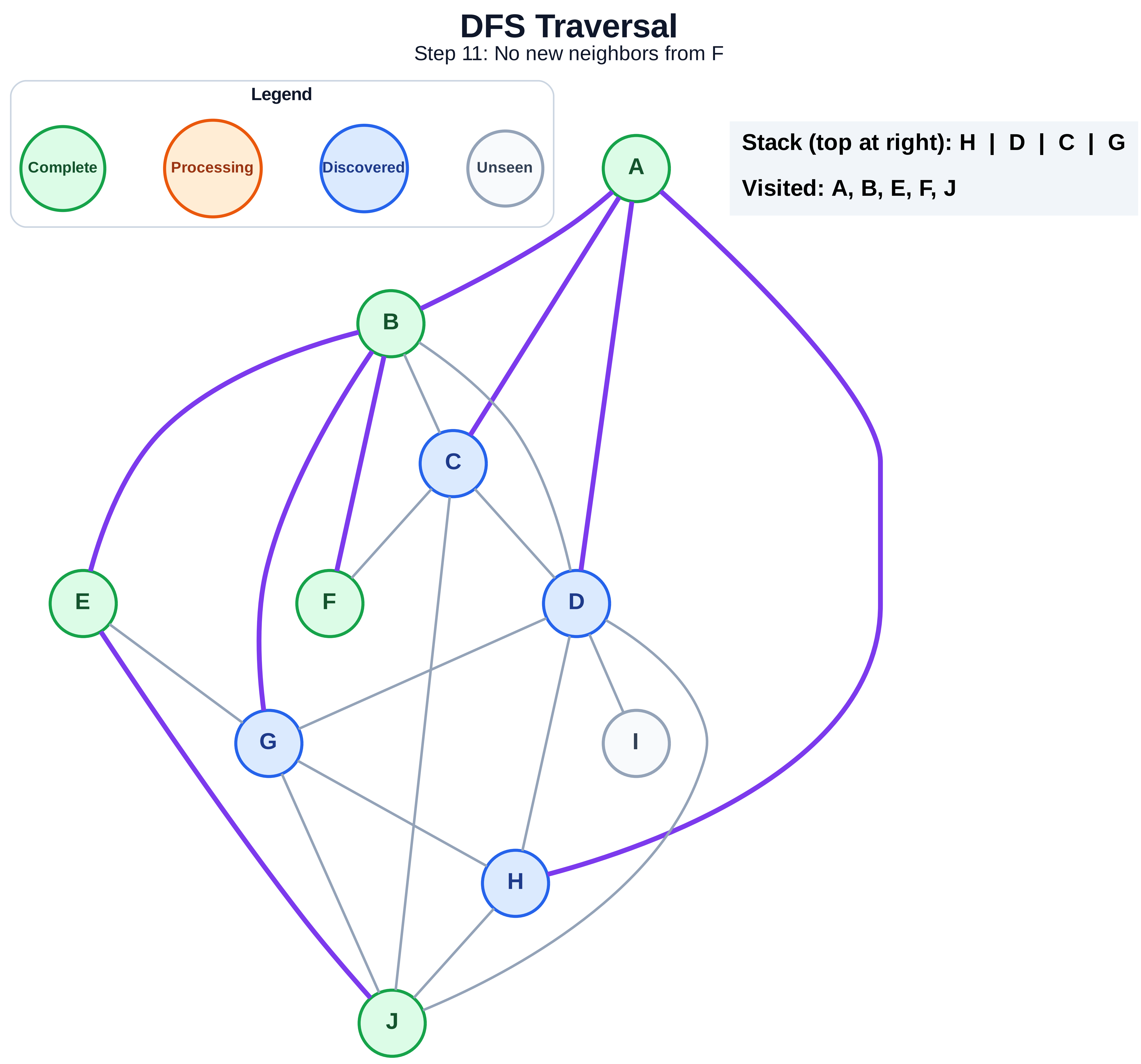
Step 9: No new neighbors from J

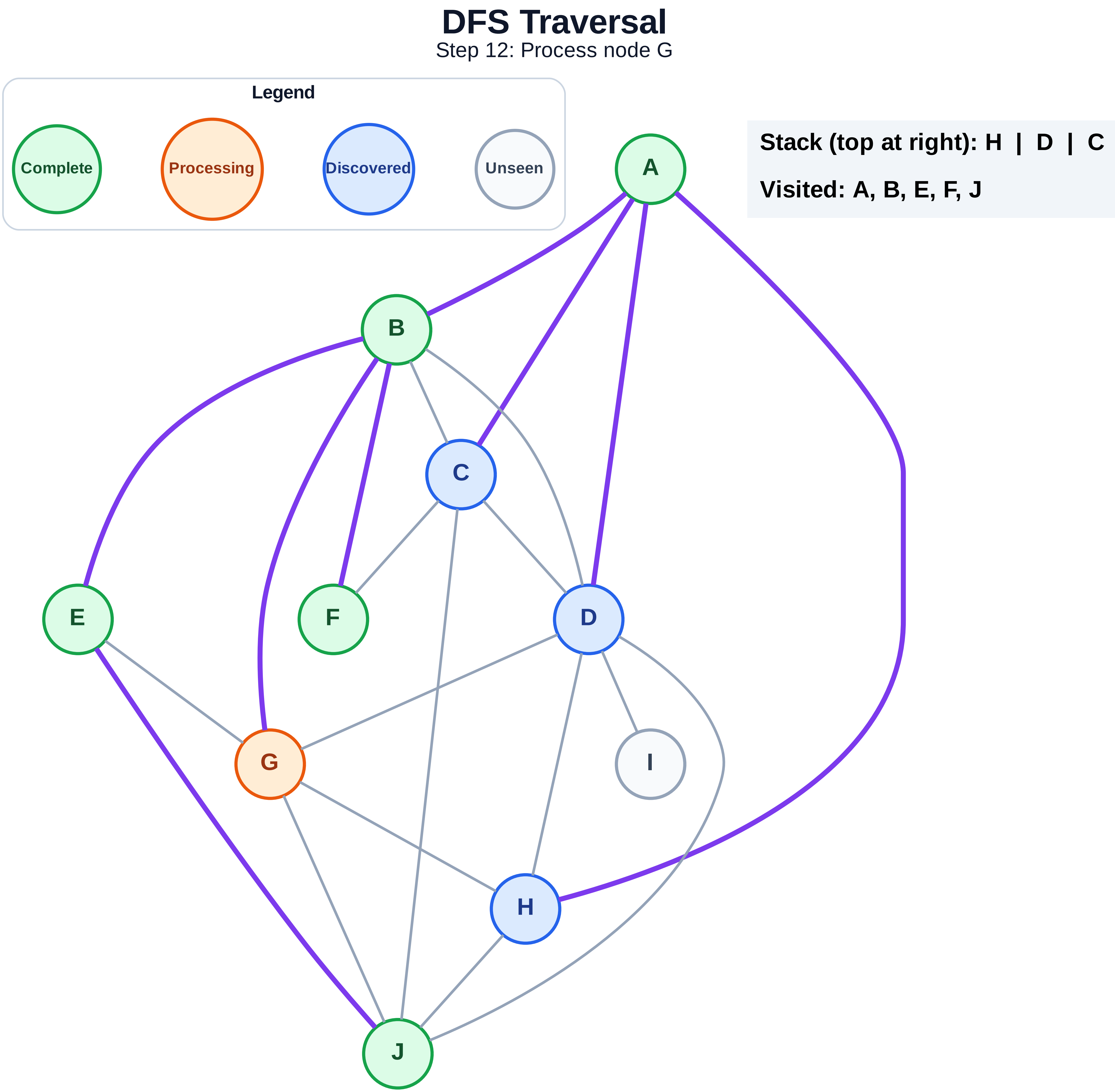


DFS Traversal

Step 10: Process node F

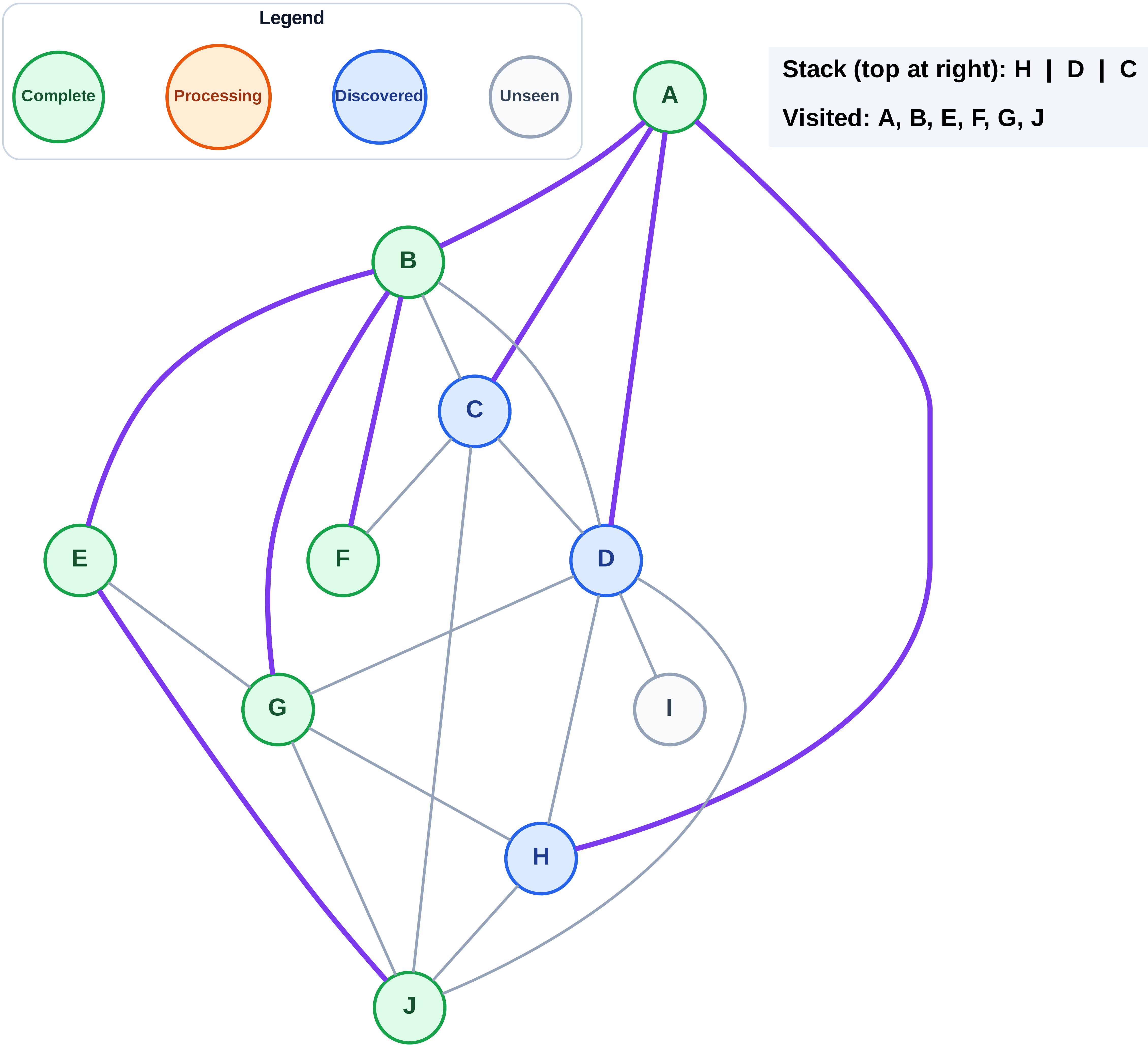



DFS Traversal
Step 11: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | D | C | G
Visited: A, B, E, F, J
A
B
C
D
E
F
G
H
I
J


DFS Traversal
Step 12: Process node G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | D | C
Visited: A, B, E, F, J
A
B
C
D
E
F
G
H
I
J

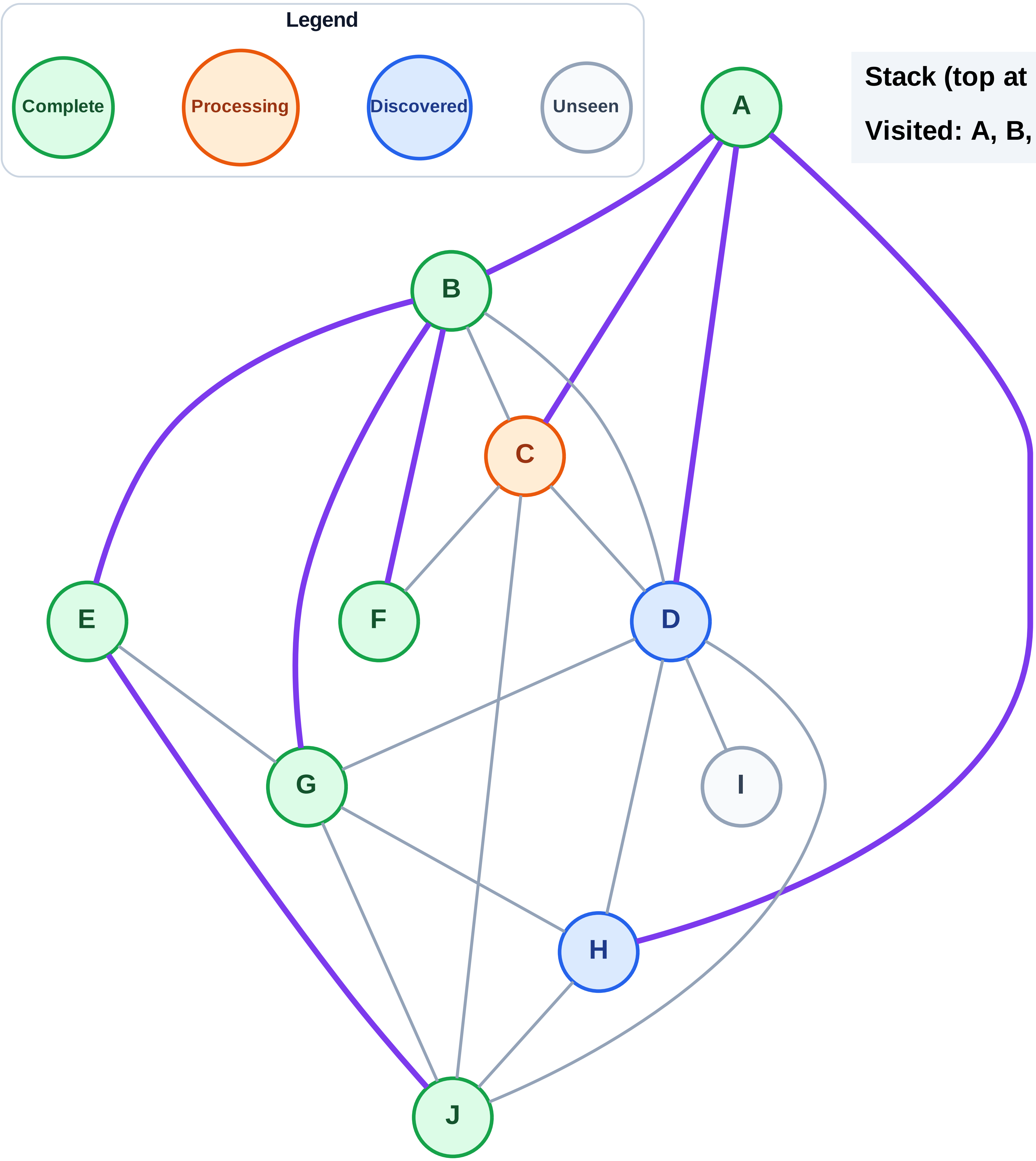
DFS Traversal

Step 13: No new neighbors from G



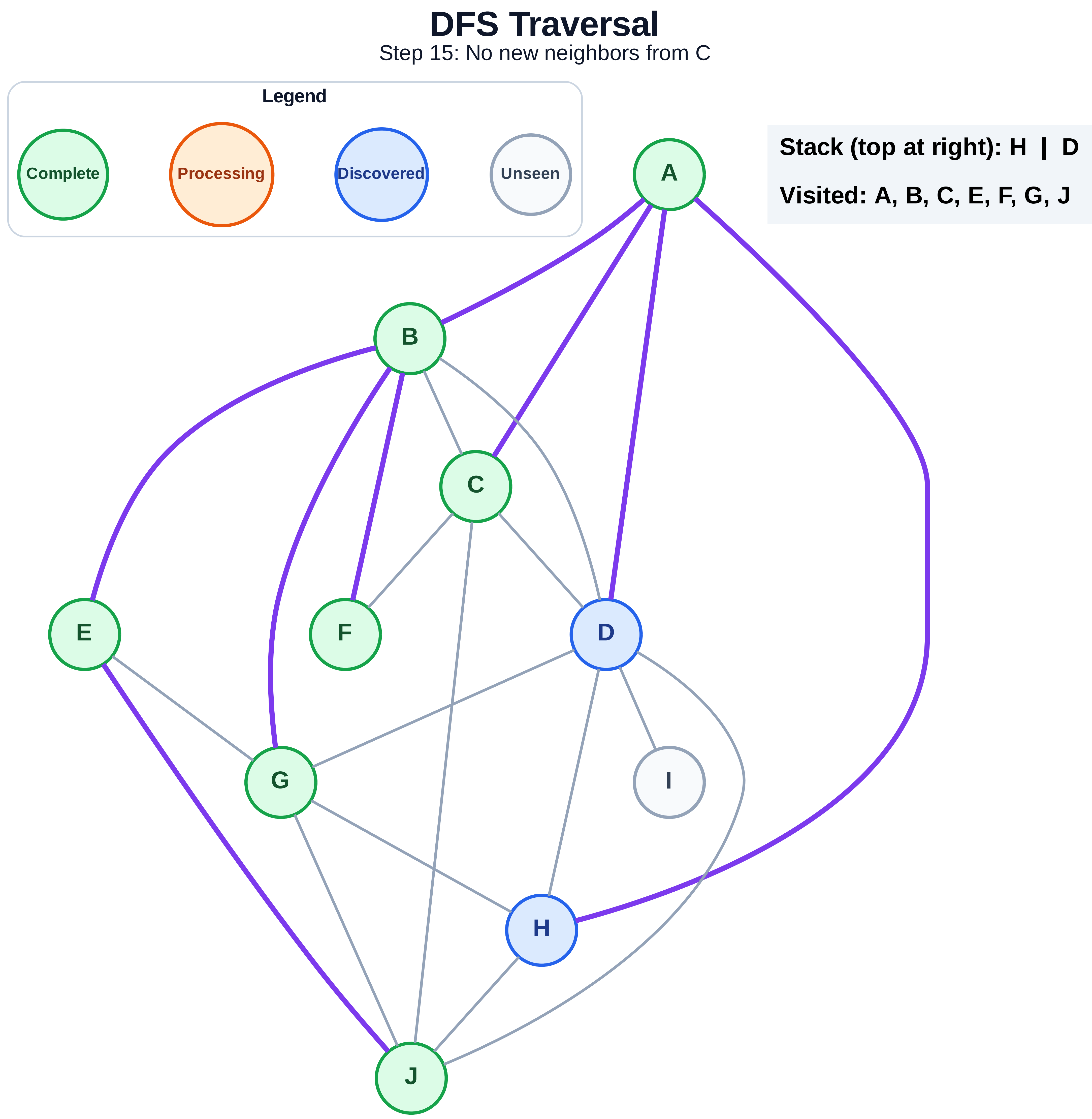
DFS Traversal

Step 14: Process node C



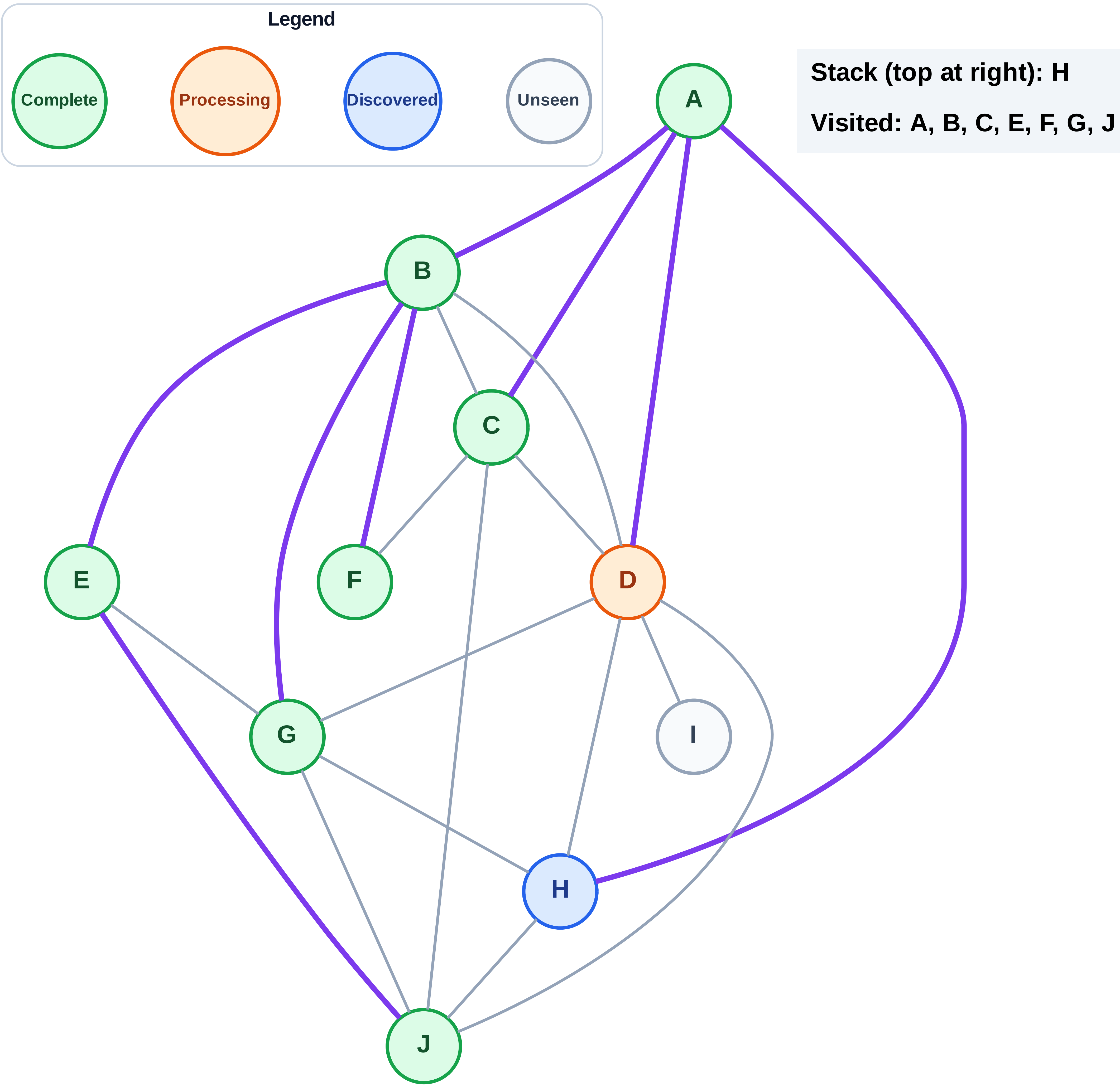
Stack (top at right): H | D

Visited: A, B, E, F, G, J


DFS Traversal
Step 15: No new neighbors from C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | D
Visited: A, B, C, E, F, G, J
A
B
C
D
E
F
G
H
I
J

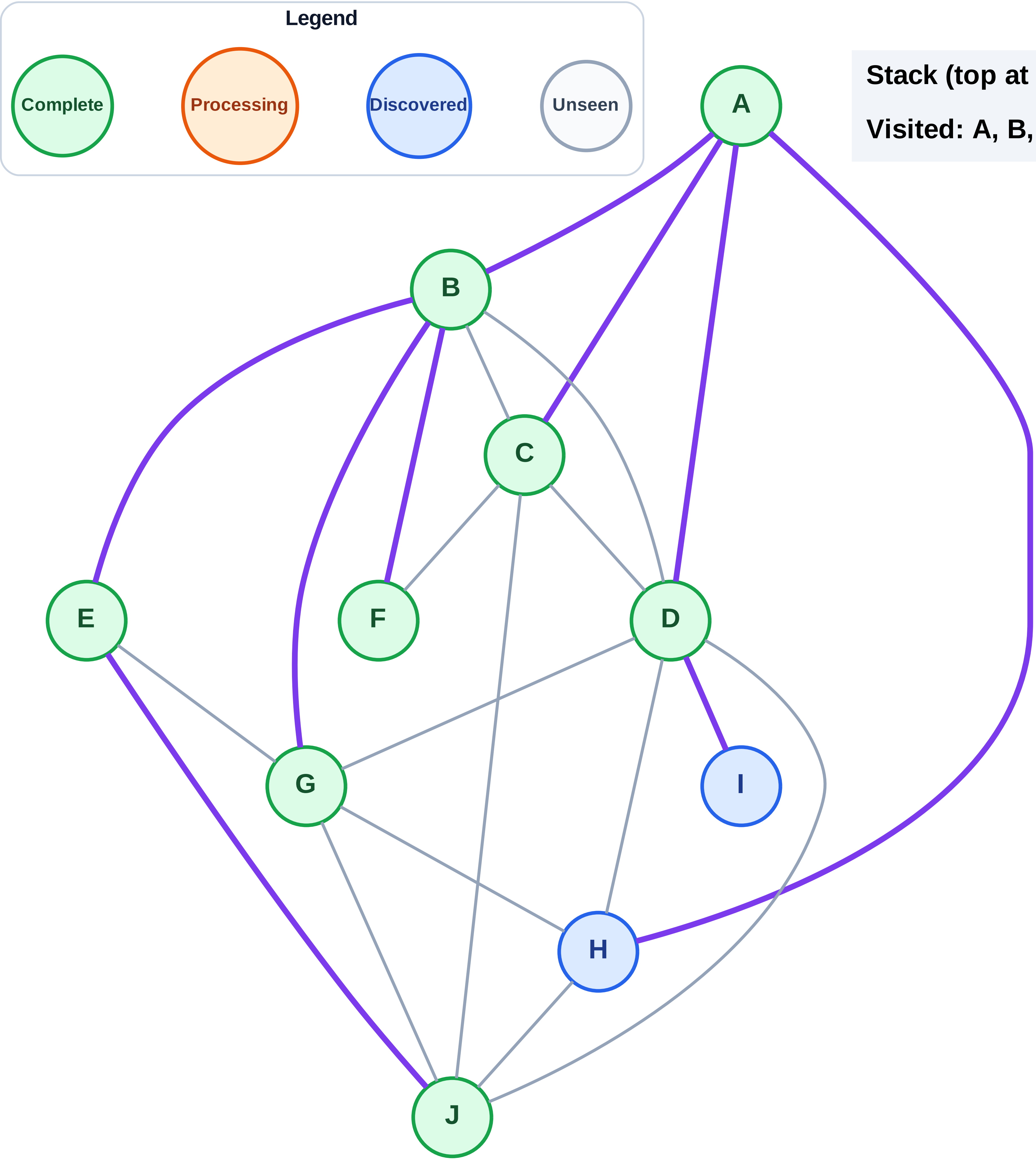
DFS Traversal

Step 16: Process node D



DFS Traversal

Step 17: Discover I from D

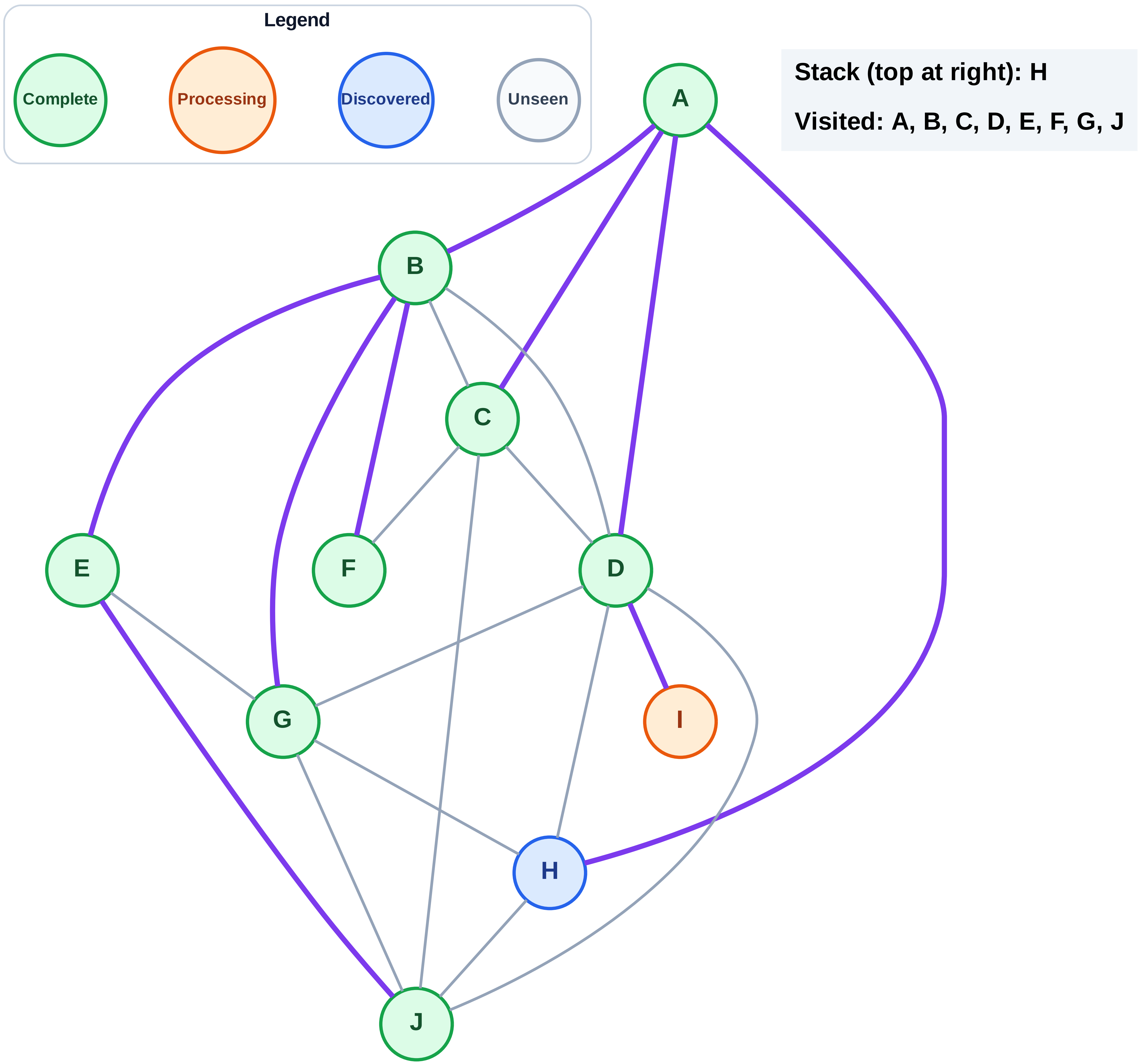


Stack (top at right): H | I

Visited: A, B, C, D, E, F, G, J

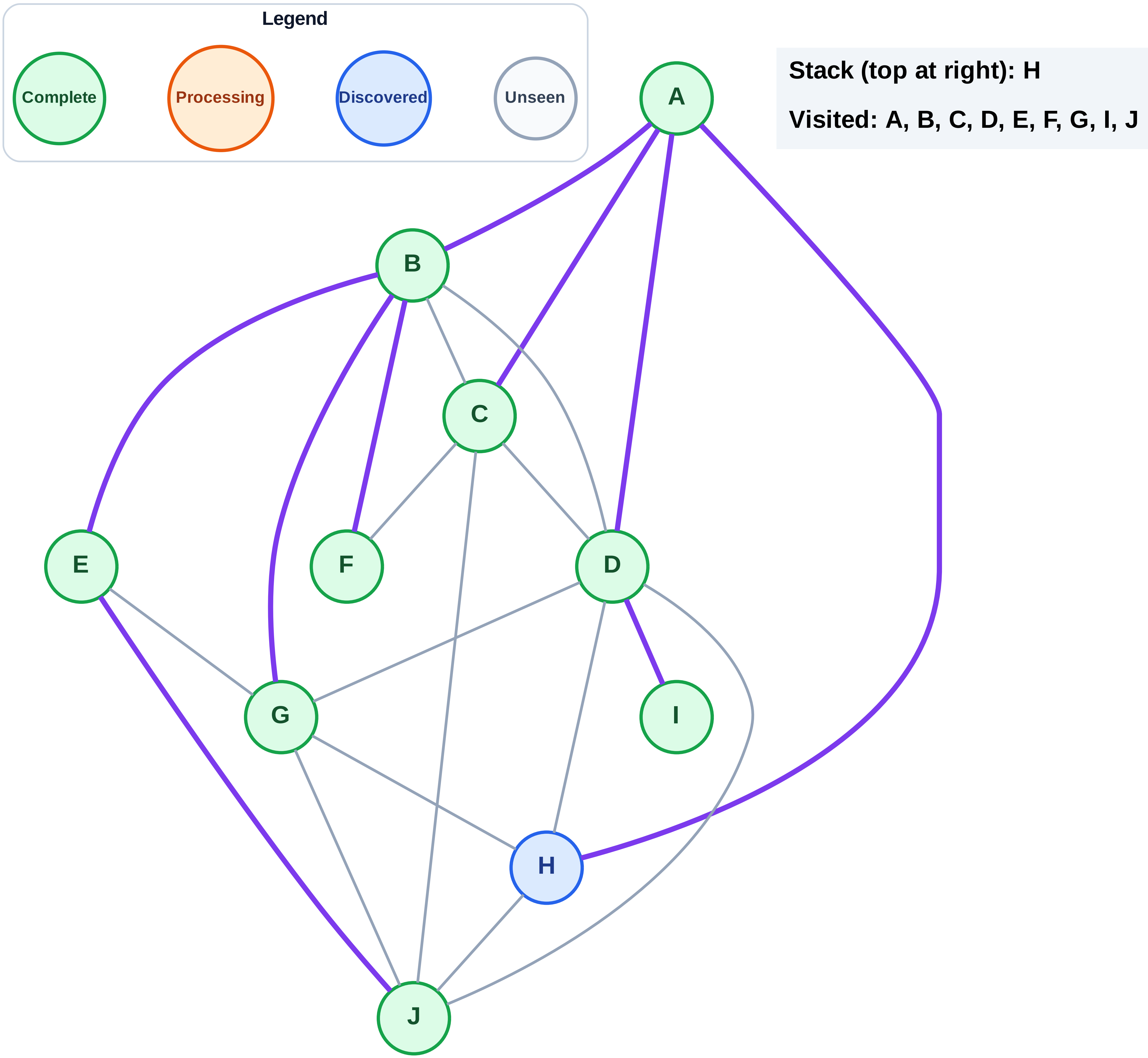
DFS Traversal

Step 18: Process node I



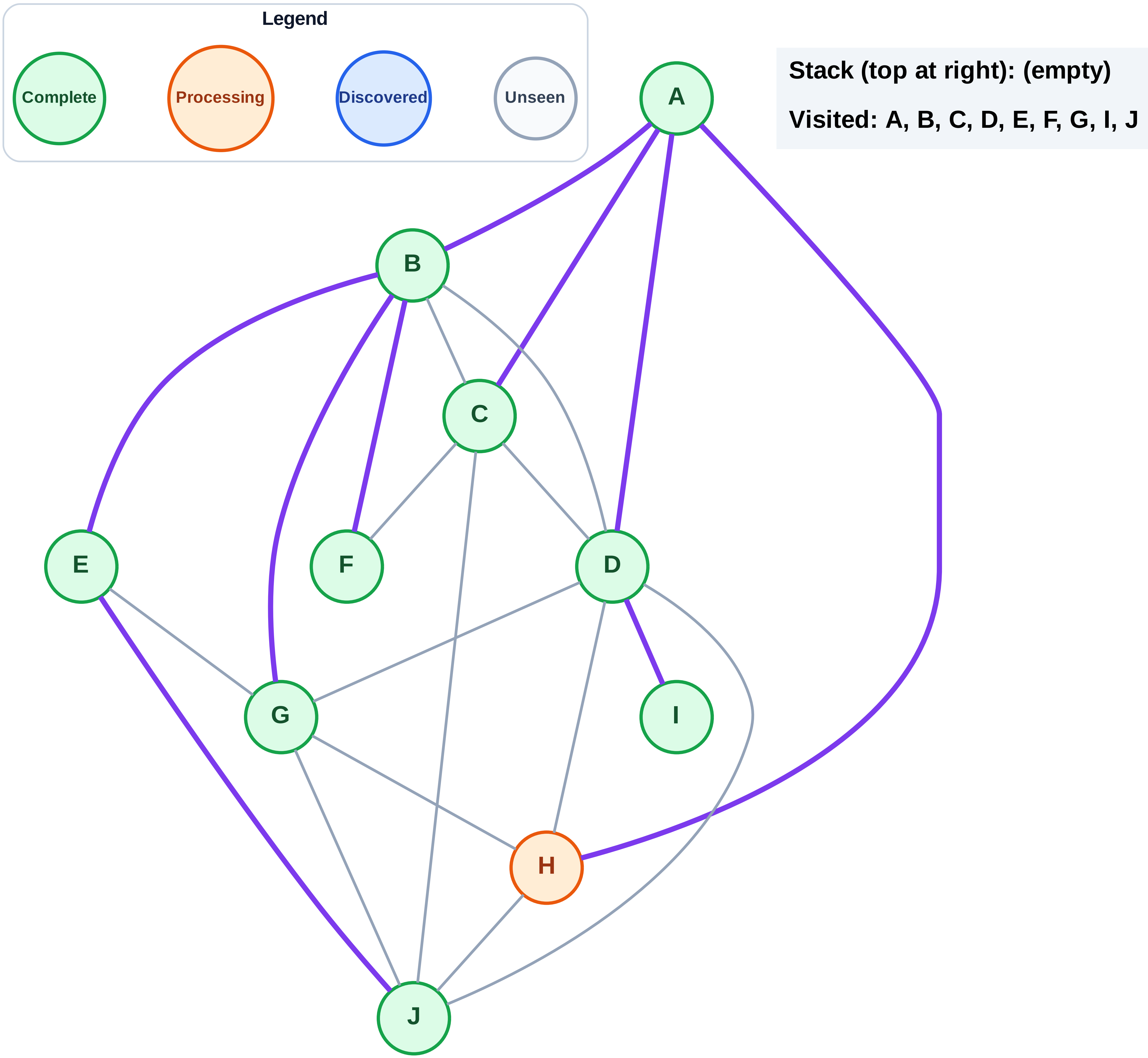
DFS Traversal

Step 19: No new neighbors from I



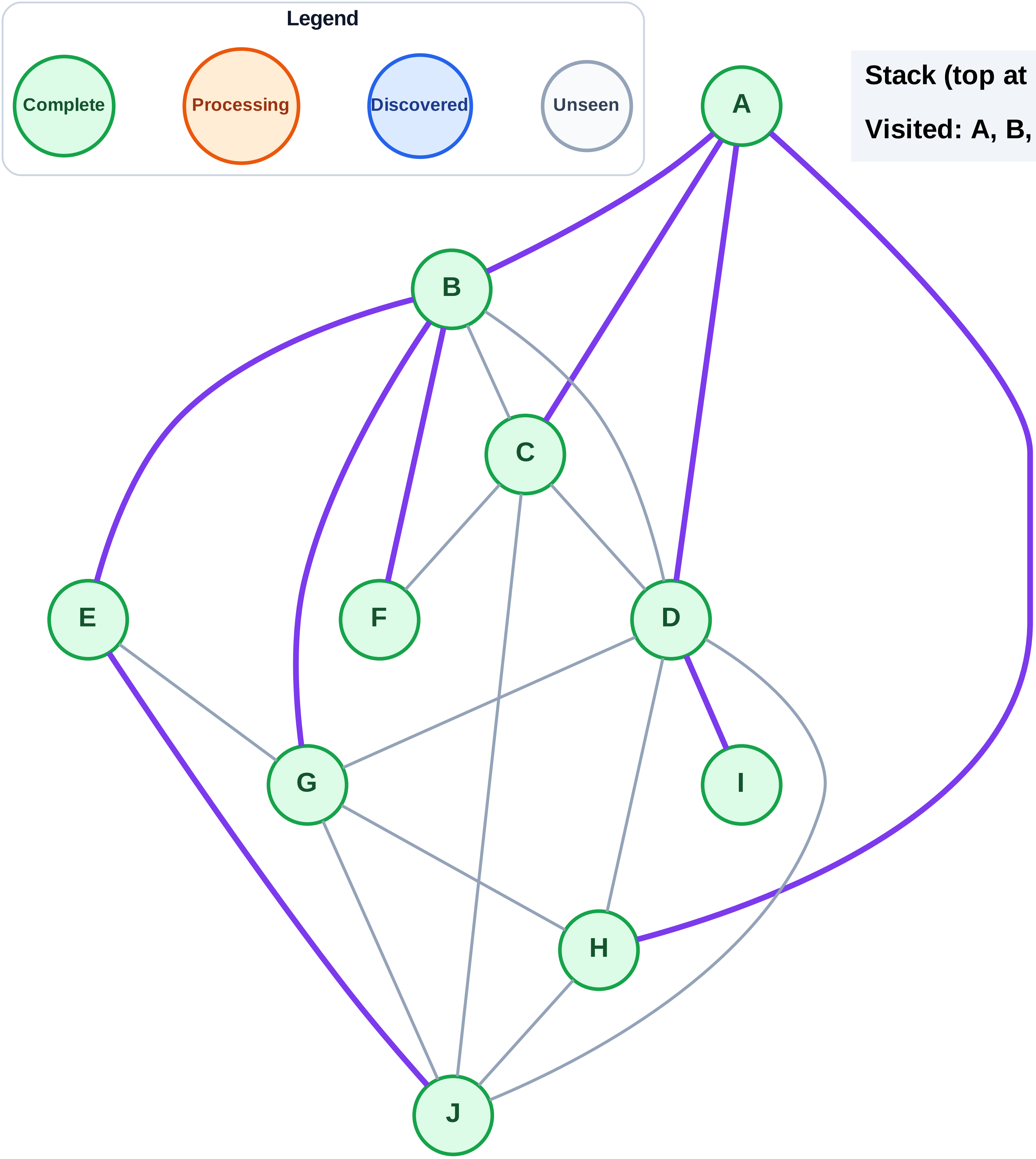
DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J